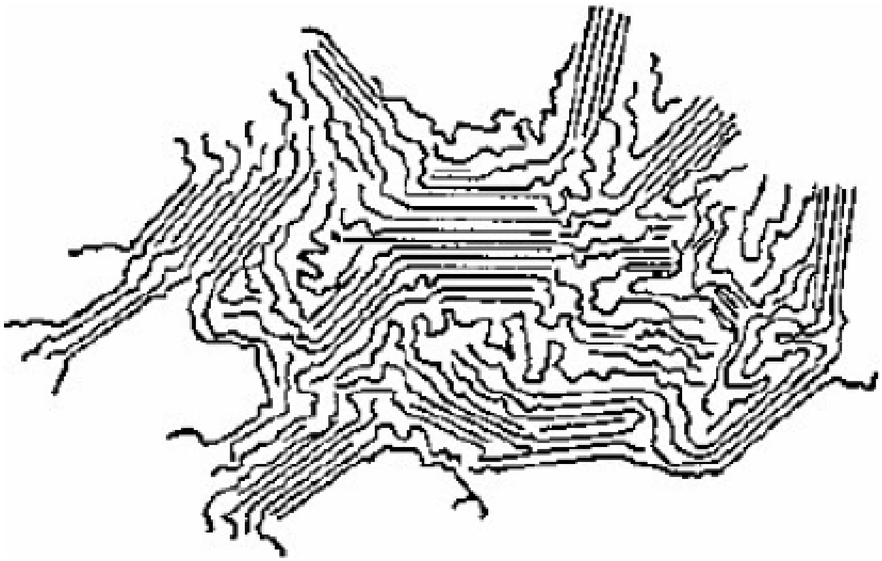
Biomaterials

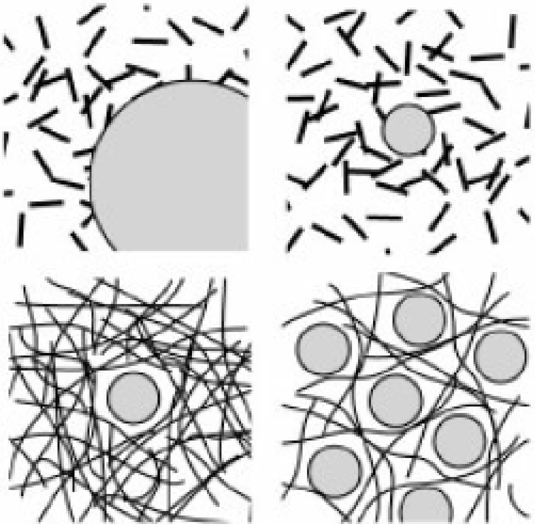
生物材料

南方科技大学材料科学与工程系2024年春季课程

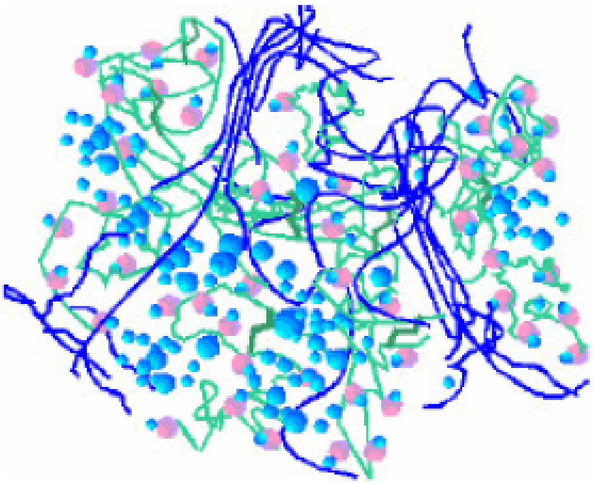
Various types of biomaterials (II)

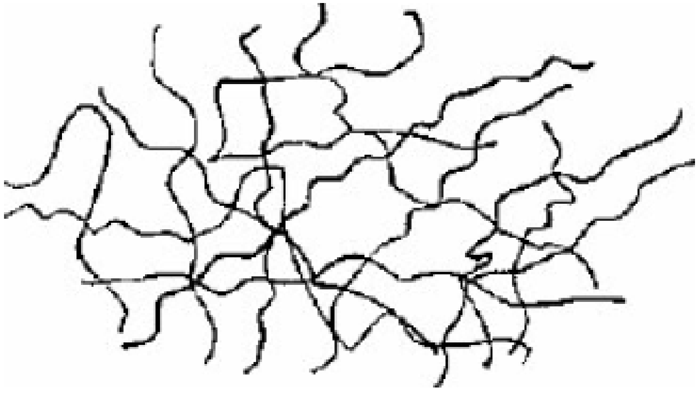
Polymeric Materials





Polymeric Materials





What is a polymer?

- The word is from Greek roots “poly” meaning many and “meros” meaning parts.

- Many scientists prefer the word “macromolecule”.

- If one discounts the end uses, the differences between all polymers, whether natural or synthetic, are determined by the intermolecular and intramolecular forces that exist between the molecules within the individual molecules and by the functional groups they contain.

Polymers

- If we disregard metals and inorganic compounds, we observe that practically everything else in the world is polymeric.

- This includes the **protein, nucleic acid and sugars that make up all cells and their extracellular matrix**, the fibers in our clothing, the food that we eat, the elastomers in our tires, the paint, plastic wall and floor coverings, our foam insulation, dishes, furniture of our homes, etc.

Polymeric Biomaterials are used in a Broad Range of Products

MEDICAL PLASTIC MARKET FORECAST TO CROSS 2.6 BILLION POUNDS BY 2004-Worldwide

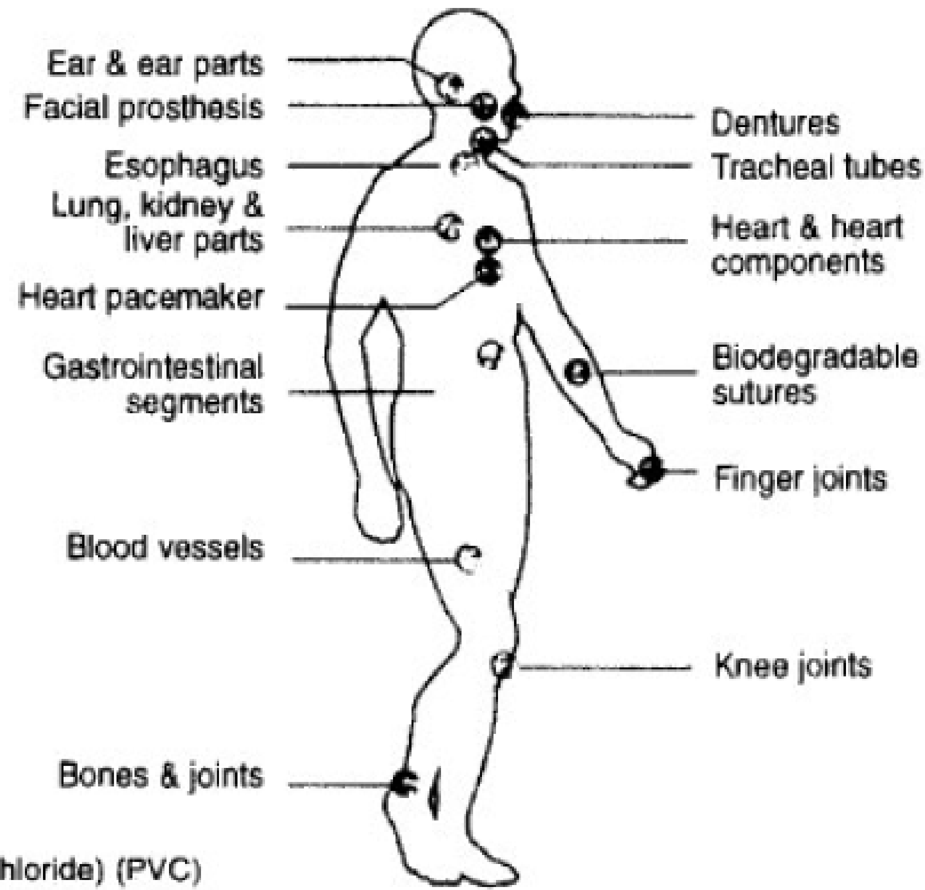
- Plastic usage in the healthcare field encompasses several distinct markets-including **disposable or single** use biomaterials.
- Predominant are applications for medical devices and related products and packaging.
- Almost 80% of polymers used in the medical industry are represented by polyvinyl chloride **(PVC), polypropylene (PP) and polystyrene (PS)**.

Medical Plastics Market

- Major nondisposable markets include testing/diagnostic equipment, surgical instruments and related equipment, prostheses/implants, dental/ophthalmic devices;

- Disposable products include syringes, kits, labware, tubing, blood bags, utensils, gloves, trays, catheters, thermometers, etc.

Common Clinical Application and Types of Polymer used in Medicine



Ear & ear parts: acrylic, polyethylene, silicone, poly(vinyl chloride) (PVC)
Dentures: acrylic, ultrahigh molecular weight polyethylene (UHMWPE), epoxy

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Dentures: acrylic, ultrahigh molecular weight polyethylene (UHMWPE), epoxy

Facial prosthesis: acrylic, PVC, polyurethane (PU, PUR)

Tracheal tubes: acrylic, silicone, nylon

Heart & heart segments: polyethylene (PE), polypropylene (PP), PVC

Blood vessels: PVC, polyester

Biodegradable sutures: PUR

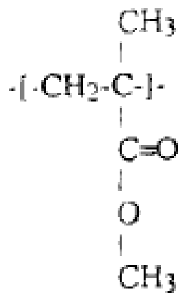
Gastrointestinal segments: Silicones, PVC, nylon

Finger joints: silicone, UHMWPE

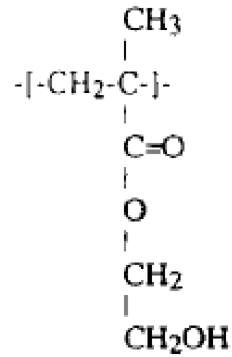
Bones & joints: acrylic, nylon, silicone, PUR, PP, UHMWPE

Knee joints: polyethylene

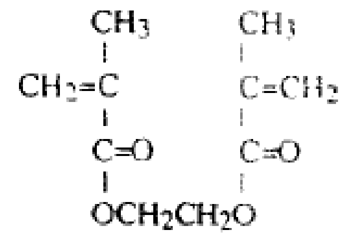
Some synthetic polymers



Poly(methyl methacrylate)
(PMMA)



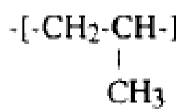
Poly(2-hydroxyethyl
methacrylate)
poly(HEMA)



Ethylene glycol
dimethacrylate
(EGDM)

$-[-CH_2-CH_2-]-$

Polyethylene
(PE)



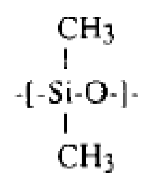
Polypropylene
(PP)

$-[-CF_2-CF_2-]-$

Poly(tetrafluoroethylene)
(PTFE)

$-[-CH_2-CHCl-]-$

Poly(vinyl chloride)
(PVC)

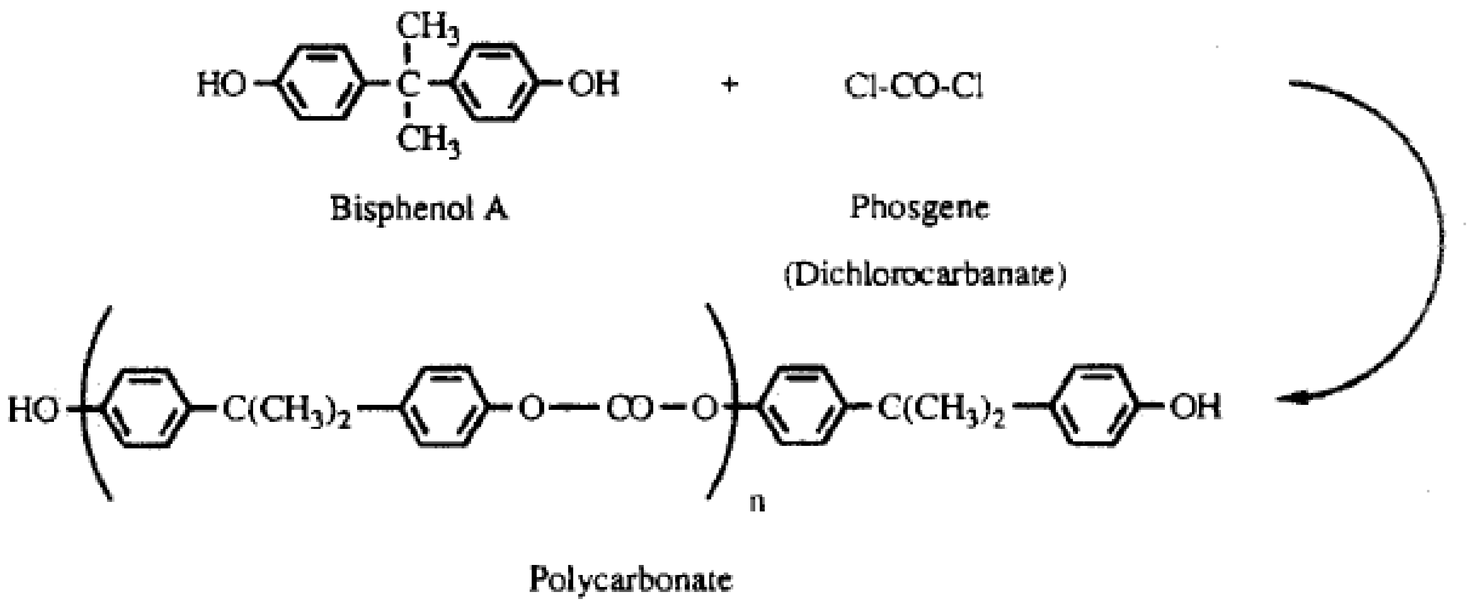


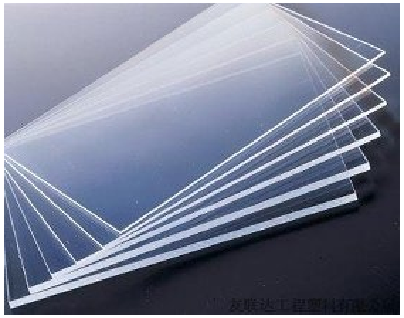
Poly(dimethyl siloxane)
(PDMS)

Polyethylene glycol

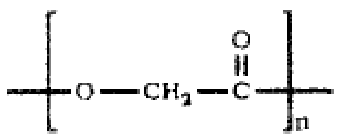
Some synthetic polymers

Polycarbonate

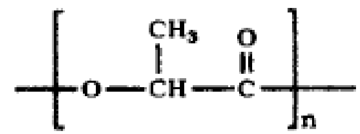




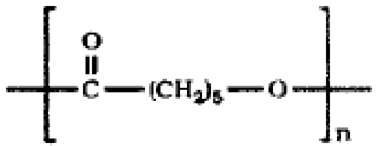
Some synthetic polymers



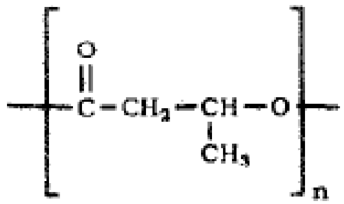
Poly(glycolic acid)



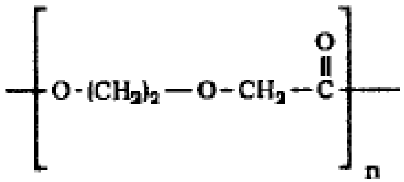
Poly(lactic acid)



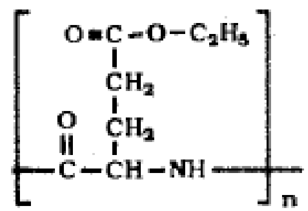
Poly(ε-caprolactone)



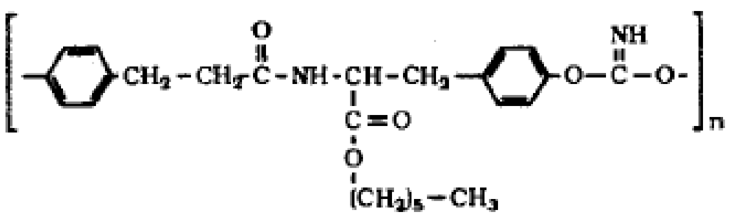
Poly(β-hydroxybutyrate)



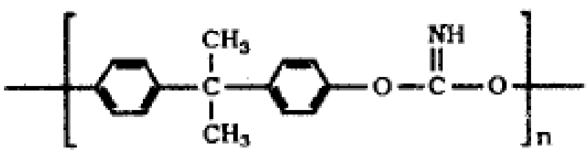
Polydioxanone



Poly(γ-ethyl glutamate)

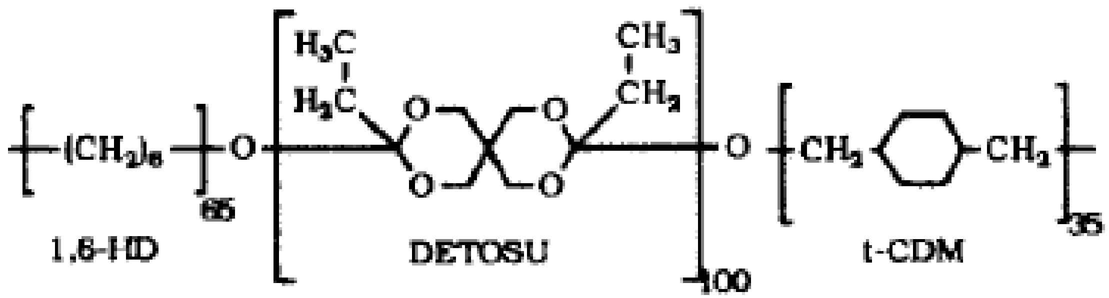


Poly(DTH iminocarbonate)

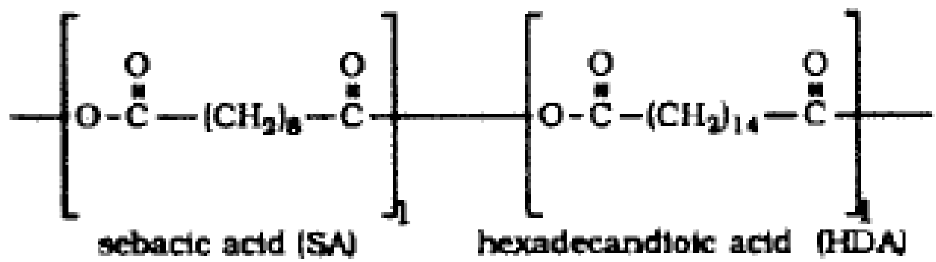


Poly(Bisphenol A iminocarbonate)

Some synthetic polymers



Poly(DETOSU- 1,6 HD-t-CDM ortho ester)

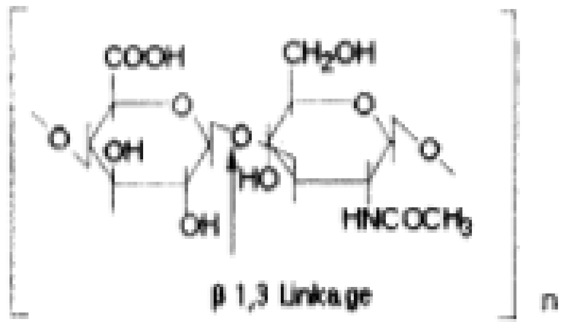


Poly(SA-HDA anhydride)

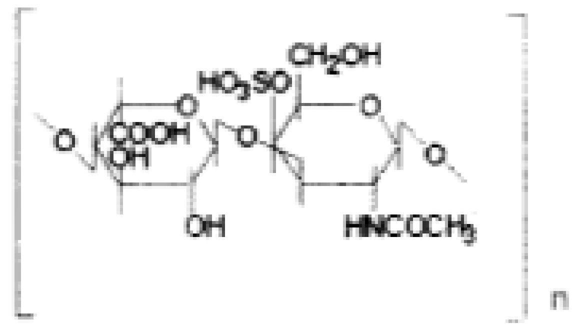
FIG. 1. Chemical structures of the degradable polymers listed in Table 2.

Nature Polymers

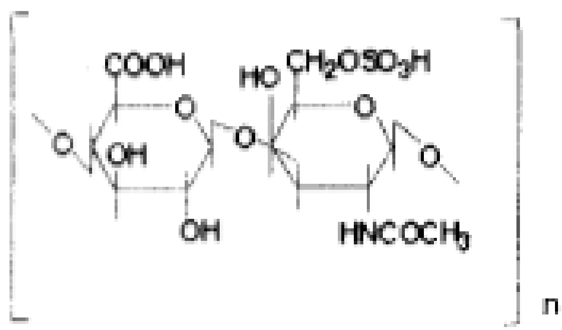
HYALURONIC ACID



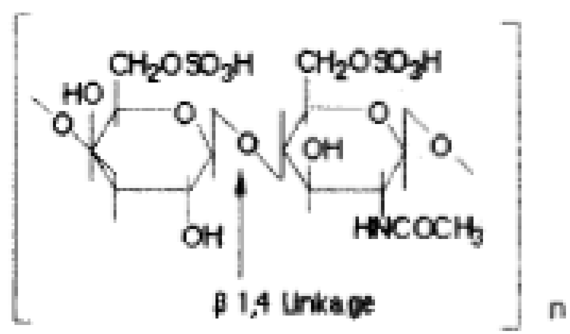
DERMATAN SULFATE



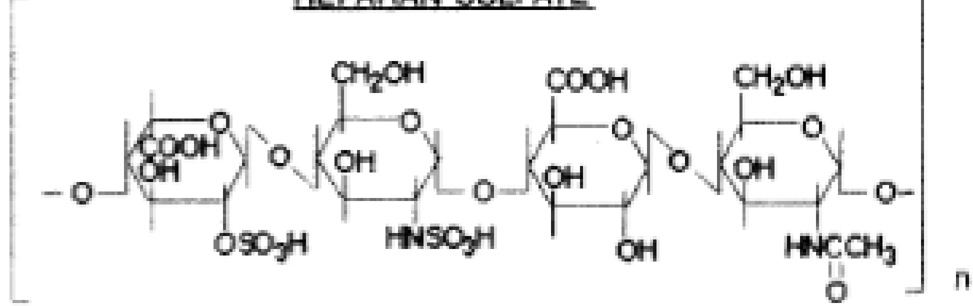
CHONDROITIN 6-SULFATE



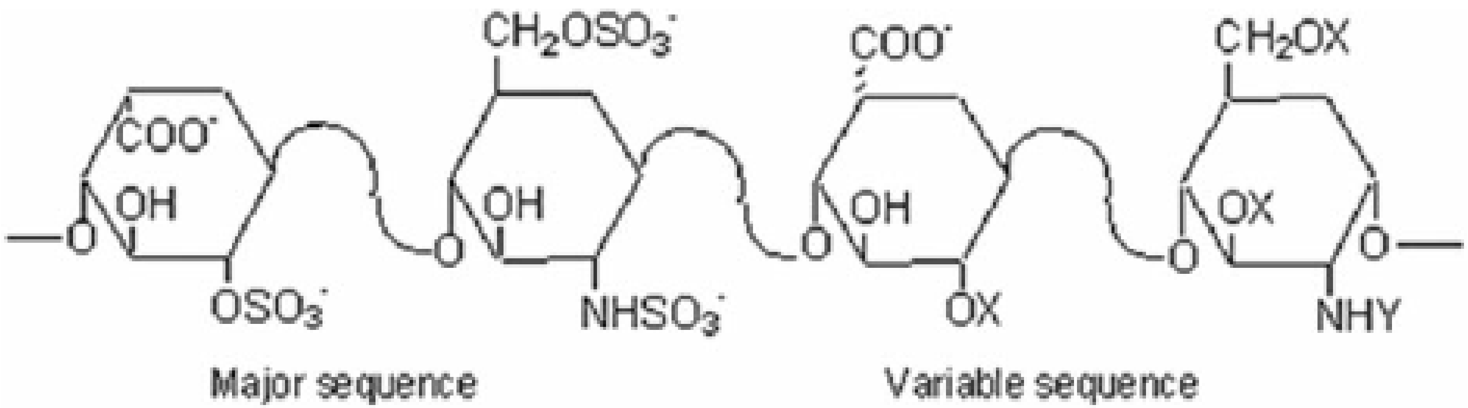
KERATAN SULFATE



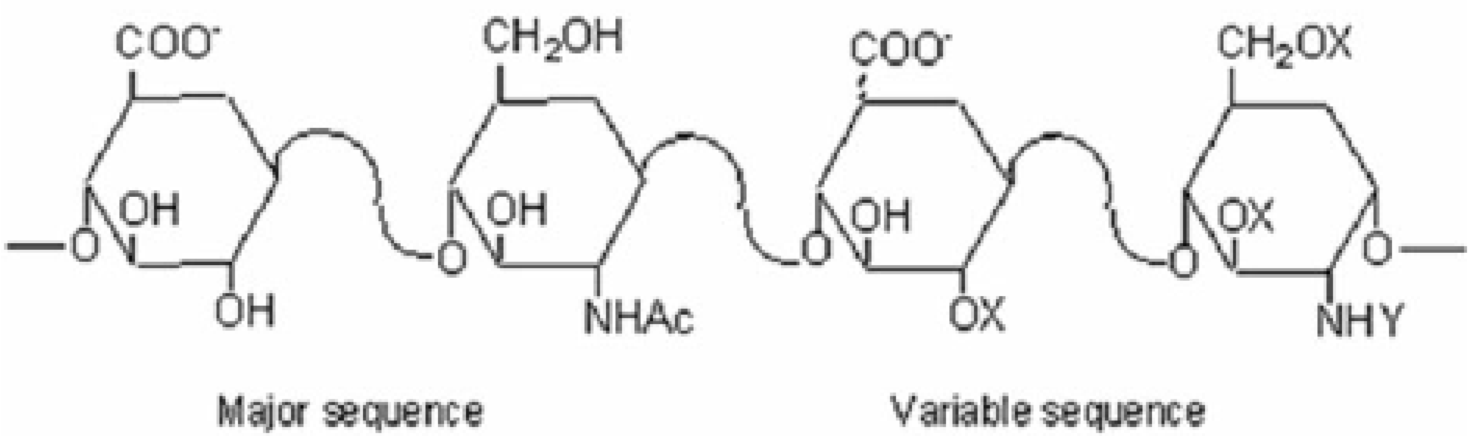
HEPARAN SULFATE



Nature Polymers



Heparin



Heparan Sulphate

Hyaluronic acid is a substance that is naturally present in the human body. It is found in the highest concentrations in fluids in the eyes and joints. The hyaluronic acid that is used as medicine is extracted from rooster combs or made by bacteria in the laboratory.

People take hyaluronic acid for various joint disorders, including osteoarthritis. It can be taken by mouth or injected into the affected joint by a healthcare professional.

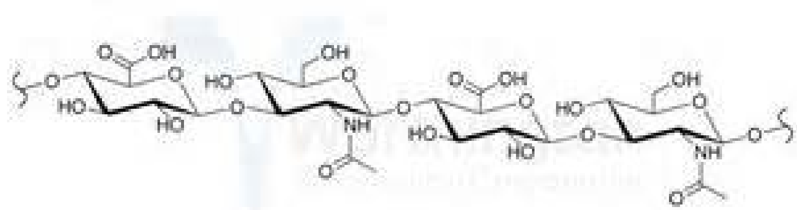
The FDA has approved the use of hyaluronic acid during certain eye surgeries including cataract removal, corneal transplantation, and repair of a detached retina and other eye injuries. It is injected into the eye during the procedure to help replace natural fluids.

Hyaluronic acid is also used as a lip filler in plastic surgery.

Some people apply hyaluronic acid to the skin for healing wounds, burns, skin ulcers, and as a moisturizer.

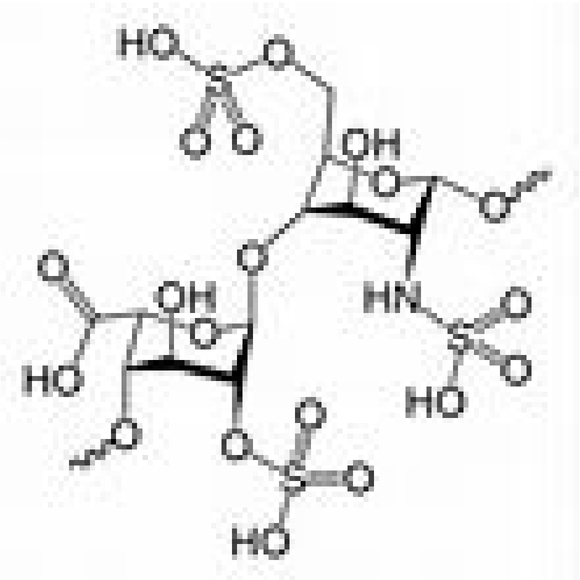
There is also a lot of interest in using hyaluronic acid to prevent the effects of aging. In fact, hyaluronic acid has been promoted as a "fountain of youth." However, there is no evidence to support the claim that taking it by mouth or applying it to the skin can prevent changes associated with aging.

Hyaluronic Acid



透明质酸

Heparan sulfate (HS) is a linear polysaccharide found in all animal tissues. It occurs as a proteoglycan in which two or three HS chains are attached in close proximity to cell surface or extracellular matrix proteins. It is in this form that HS binds to a variety of protein ligands and regulates a wide variety of biological activities, including developmental processes, angiogenesis, blood coagulation, abolishing detachment activity by GrB, and tumor metastasis. HS has been shown to serve as cellular receptor for a number of viruses including the respiratory syncytial virus (合胞病毒).

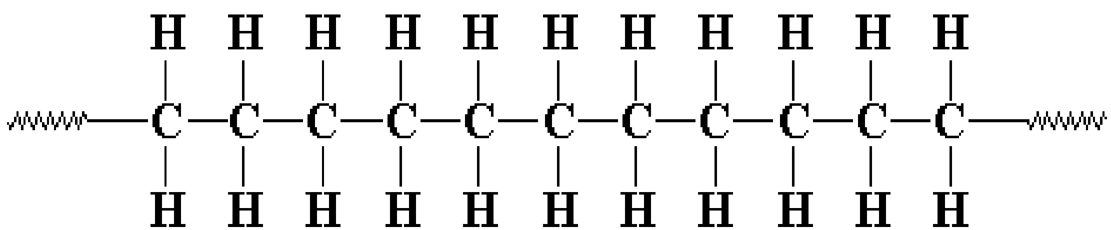


硫酸乙酰肝素

GrB: growth factor receptor-bound protein

The Structures of Polymers

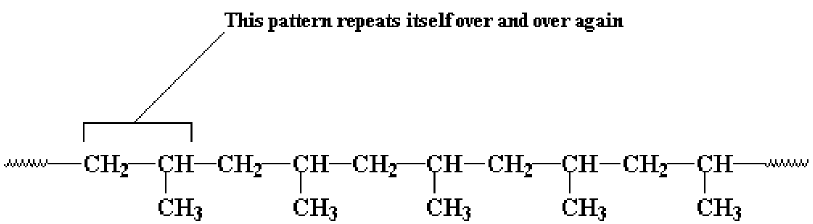
- Below is a diagram of polyethylene, the simplest polymer structure



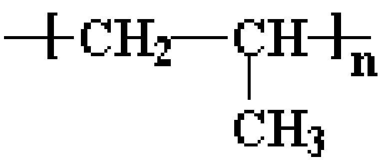
- There are polymers that contain only carbon and hydrogen.
- These are referred to as hydrocarbons-exs. Polypropylene, polybutylene, polystyrene, and polymethylpentene

Polymers have a Repeating Structure

- We like to think that the atoms that make up the backbone of a polymer chain come in a regular order, and this order repeats itself all along the length of the polymer chain.
- For example, in polypropylene, the backbone chain is made up of just two carbon atoms repeated over and over again.



Polypropylene



Basics of Polymer Structure

- What distinguishes polymers from other organic compounds is molecular weight and dimension?

TABLE 1.1 Properties of the Alkane Series

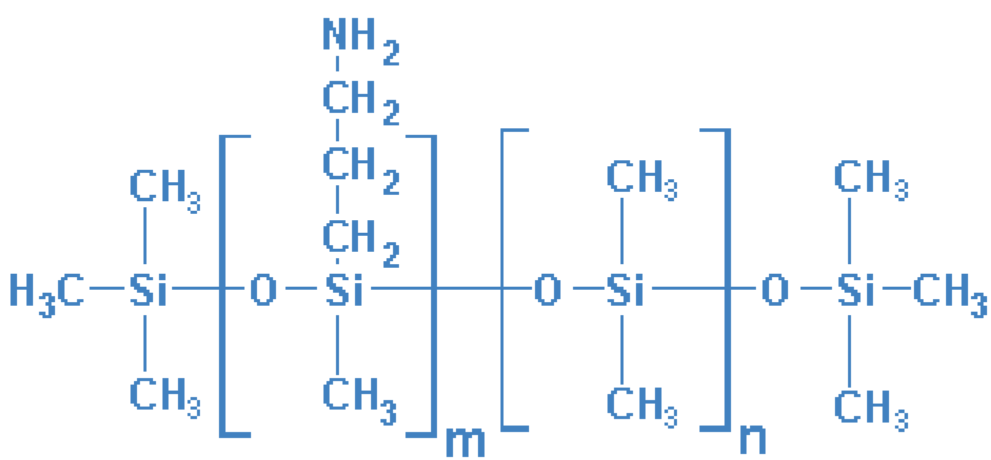
Number of Carbons in Chain	State and Properties of Material	Use, Dependent on Chain Length
1–4	Simple gas	Bottled gas for cooking
5–11	Simple liquid	Gasoline
9–16	Medium-viscosity liquid	Kerosene
16–25	High-viscosity liquid	Oil and grease
25–50	Simple solid	Paraffin wax candles
1000–3000	Tough plastic solid	Polyethylene bottles and containers

The Structures of Polymers

- Even though the basic makeup of many polymers is carbon and hydrogen, other elements can also be involved.
- Oxygen, chorine, fluorine, nitrogen, silicon, phosphorous, and sulfur are other elements found in the molecular makeup of polymers.
- Polyvinyl chloride (PVC) contains chlorine.
- Nylon contains nitrogen.
- Teflon contains fluorine.
- Polyester and polycarbonates contain oxygen.

The Structure of Polymers

- There are also some polymers that, instead of having a carbon backbone, have a silicon or phosphorous backbone.
- These are considered inorganic polymers.
- Polysiloxanes (Silicones) and Polyphosphazenes



Kelo-Cote (Polysiloxanes/Silicon Dioxide) can be used to treat scars by reducing pain, itching, and discomfort, whilst also reducing the scar`s appearance making it less noticeable. It is a silicone based gel which is self-drying, odorless, and transparent when applied.

The Structure of Polymers

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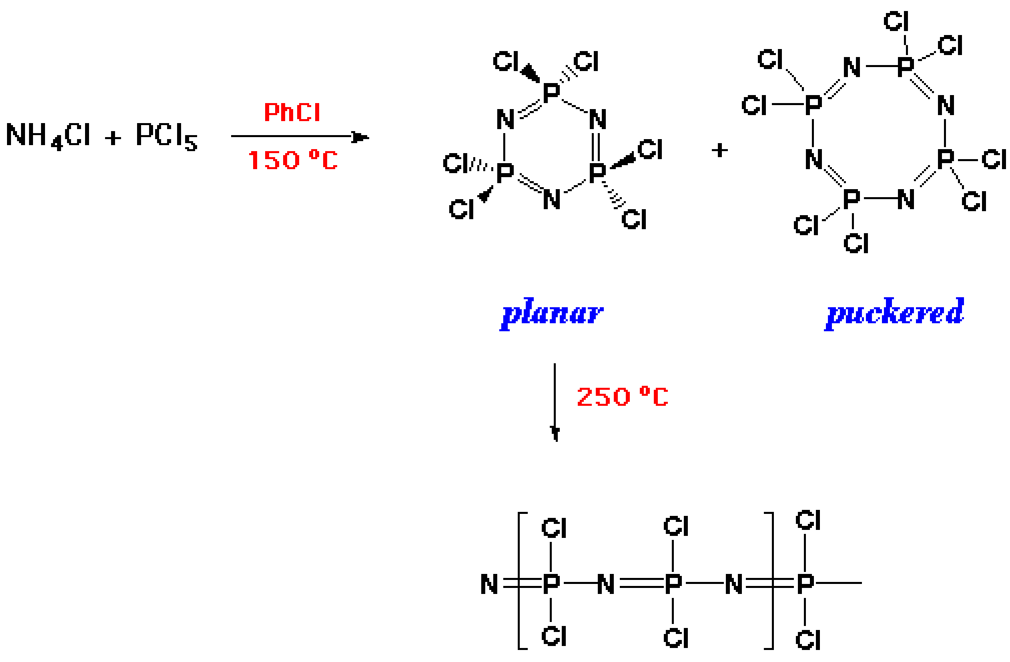
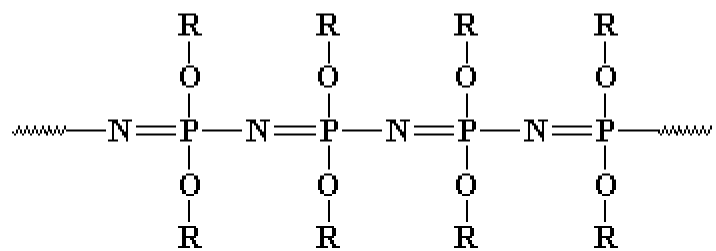


Polyphozphazene

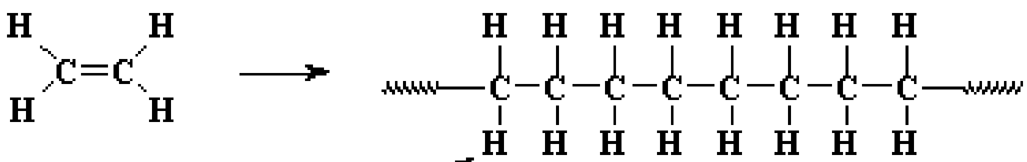


An ether substituted polyphosphazene. (*R* is a wild card, standing for any hydrocrabon group)

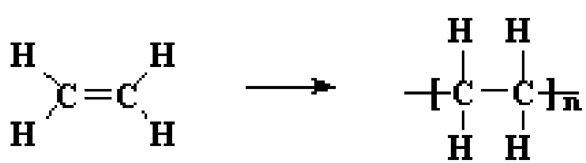
Vinyl Polymers

- Vinyl polymers are polymers made from *vinyl monomers*; that is, small molecules containing carbon-carbon double bonds.
- They make up largest family of polymers.
- Let's see how we get from a vinyl monomer to a vinyl polymer using for an example the simplest vinyl polymer, polyethylene.

Polyethylene



This can get tedious to draw, so
we often use shorthand like this.

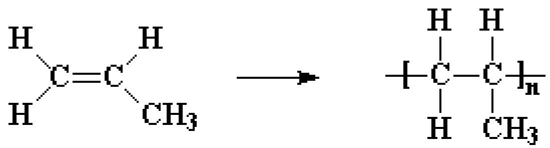


(Note: A line drawn between two atoms represents a pair of electrons shared by those atoms, which constitutes a chemical bond. Two lines represent two pairs of shared electrons, a double bond.)

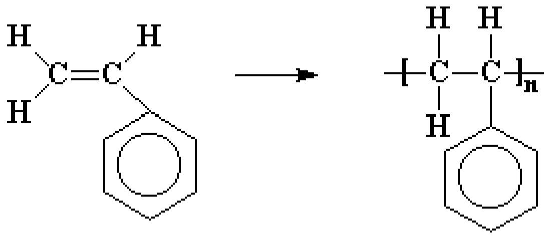
And when we're feeling really lazy we just draw it like this:

$$CH_2{=}CH_2 \longrightarrow -\!\!\left[CH_2-CH_2\right]\!\!-_n$$

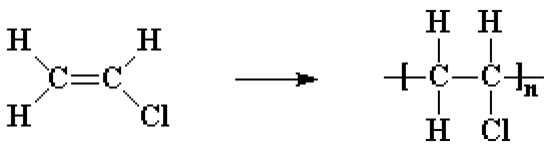
Vinyl Polymers



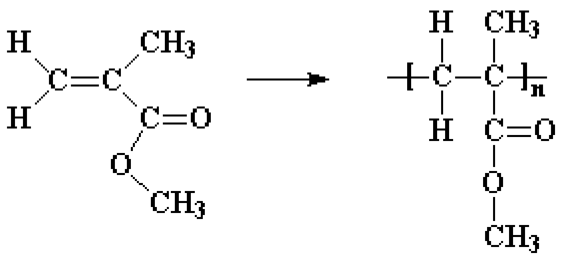
polypropylene



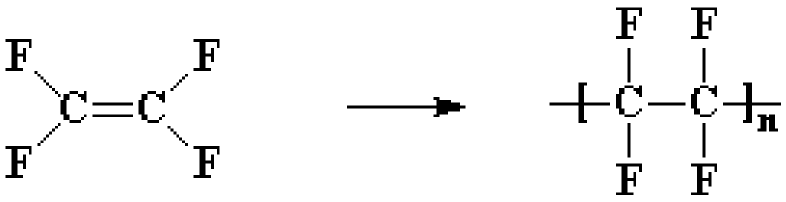
polystyrene



polyvinylchloride



polymethylmethacrylate

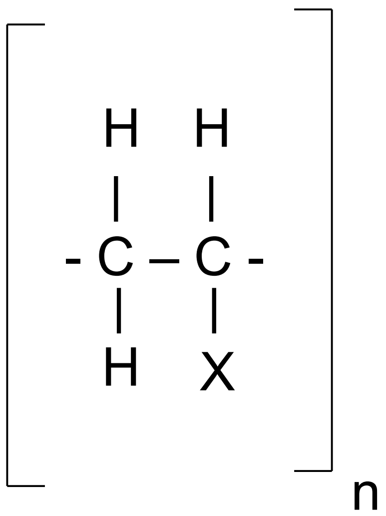


Polytetraflouroethylene (PTEF)

Vinyl Polymers

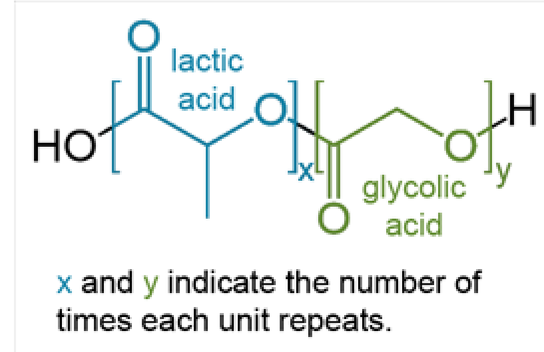
Homopolymer

- If X=H then polyethylene
- If X = CH3 then polypropylene
- If X = Cl then polyvinylchloride
- If X = Benzene ring then polystyrene

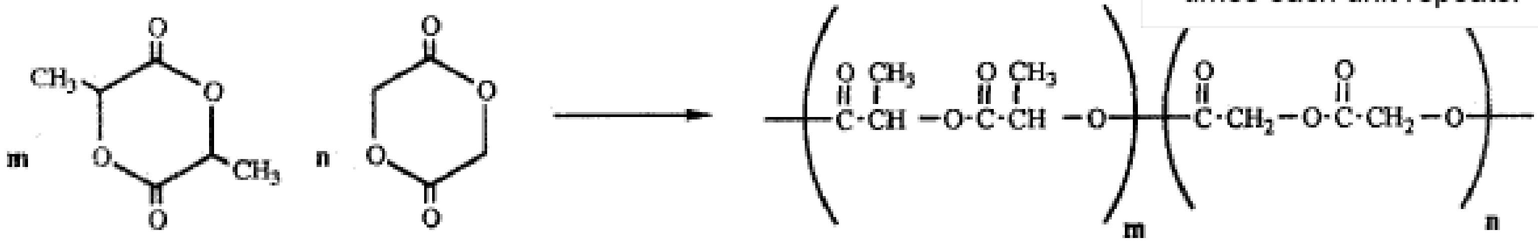


Copolymers??

Polyesters



Poly(glycolide-lactide) copolymer



Polyurethane

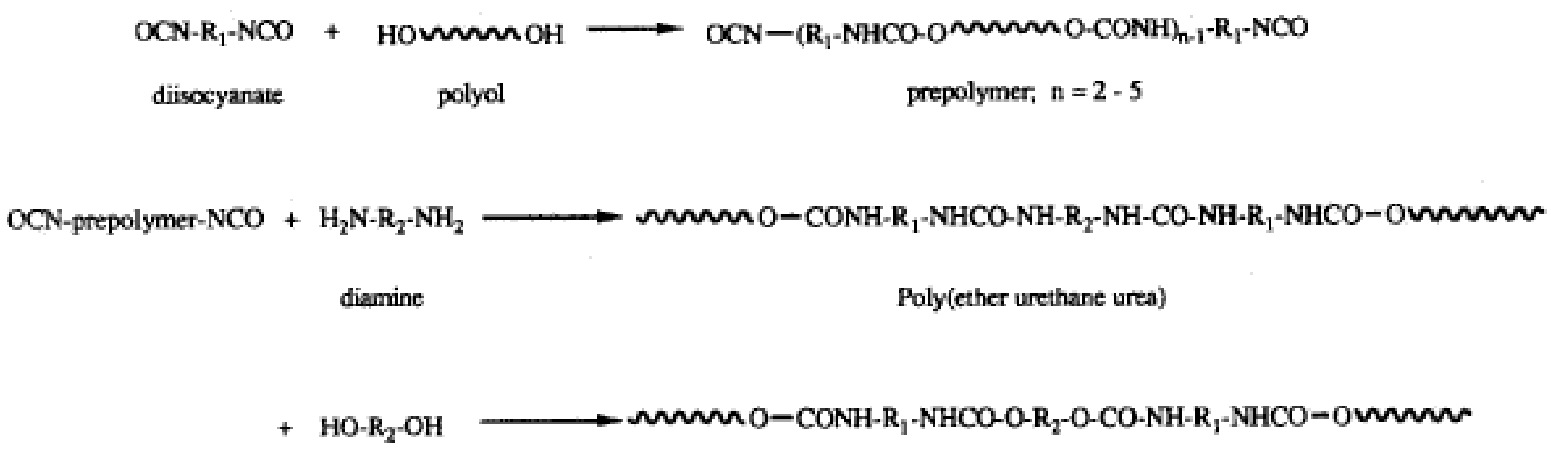
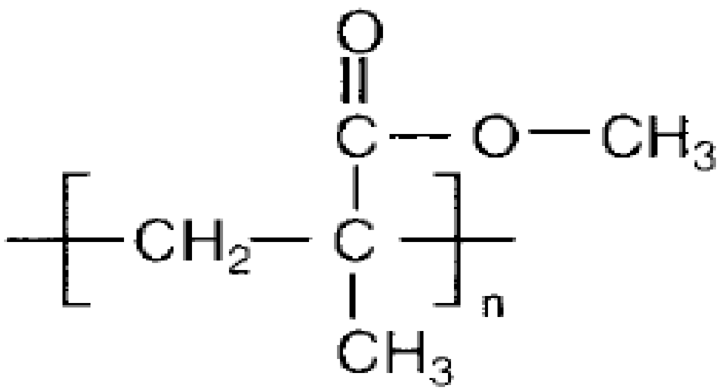
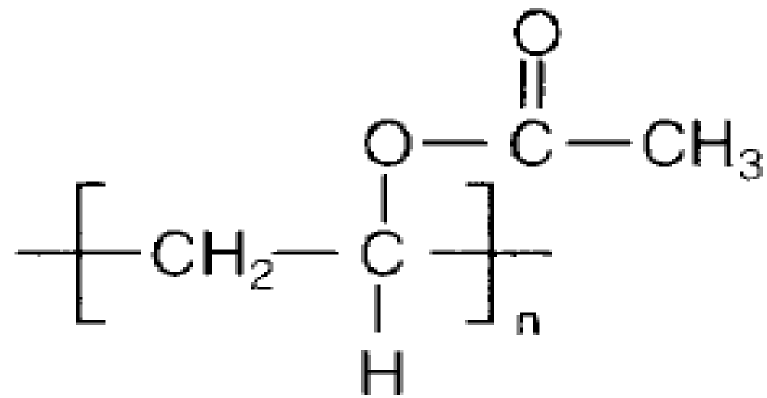


FIG. 15. Copolymers and their base monomers used in medicine.

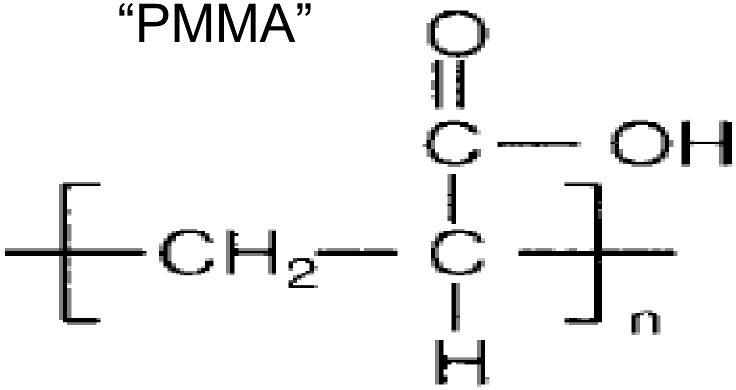
Chemical Structures of Some Common Polymers



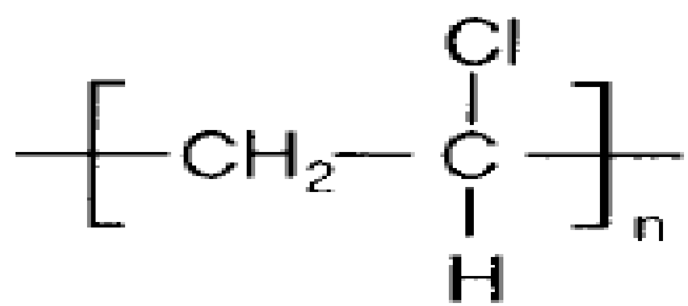
Poly(methylmethacrylate) “PMMA”



Poly(vinylacetate) “PAVc”

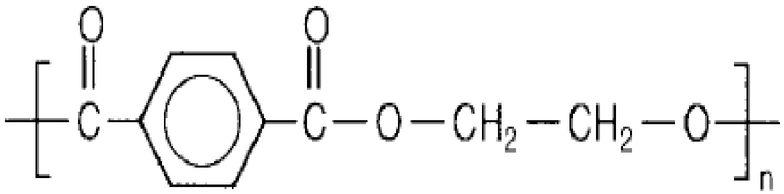


Poly(acrylate) “PAA”

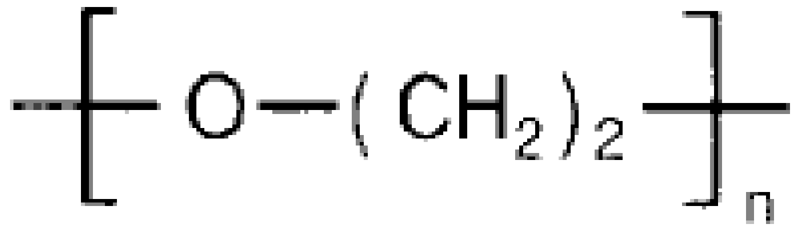


Poly(vinylchloride) “PVC”

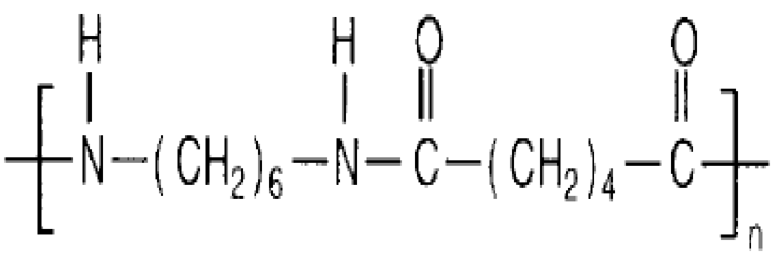
Chemical Structures of Some Common Polymers



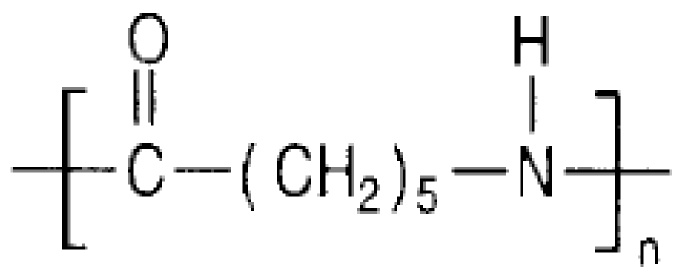
Poly(ethylene terephthalate)“PET”



Poly(ethylene oxide)“PEO”



Poly(hexamethylene adipamide)
“Nylon 6,6”



Poly(caprolactam) “Nylon”

Classification- Chain Architecture: Linear Structures

- Many thermoplastic polymers are built so their molecules consist of many thousands of atoms arranged into long linear chains. But they don't have to be long straight chains.

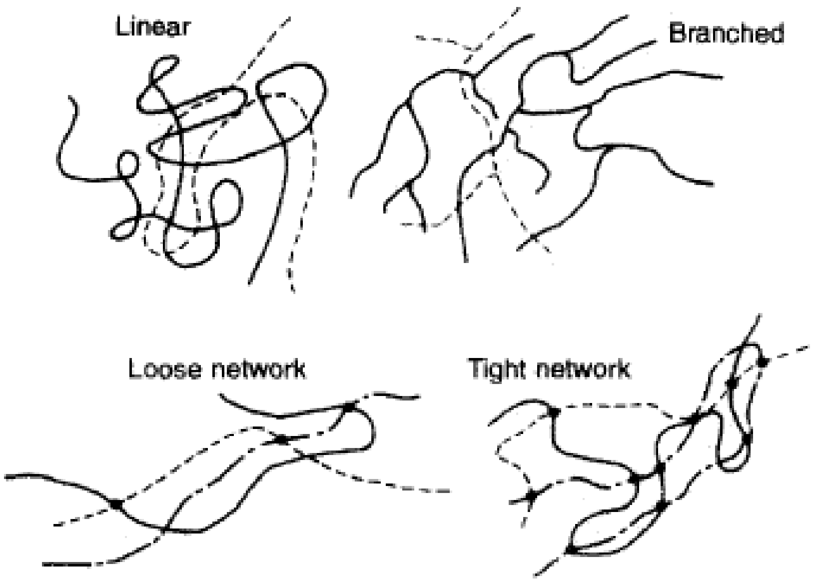


FIG. 2. Polymer arrangements. (From F. Rodriguez, *Principles of Polymer Systems*, Hemisphere Publ., 1982, p. 21, with permission.)

Branched Polymers

- Not all polymers are linear in this way. Sometimes there are chains attached to the backbone chain which are comparable in length to that backbone chain.
- Some thermoplastic polymers, like polyethylene, can be made in linear or branched versions.
- This gives them a 2-D quality.

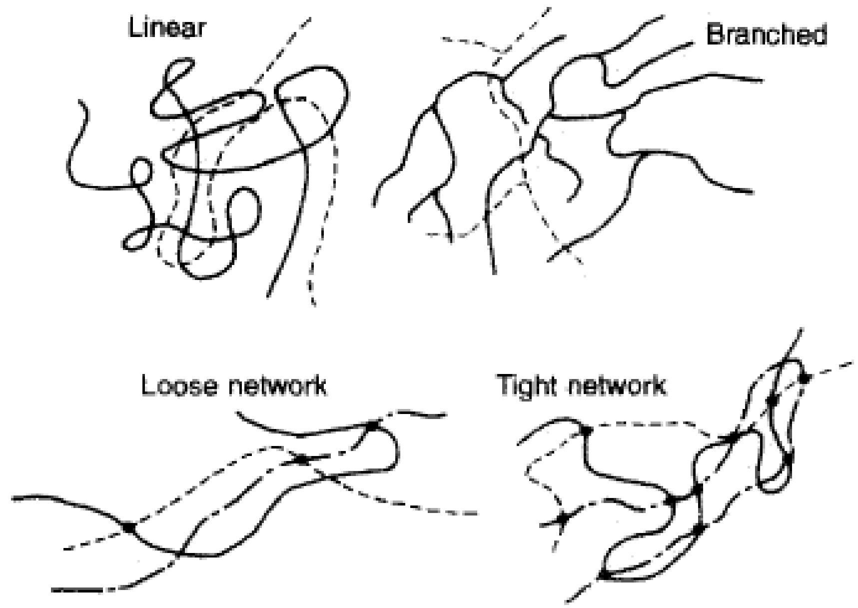
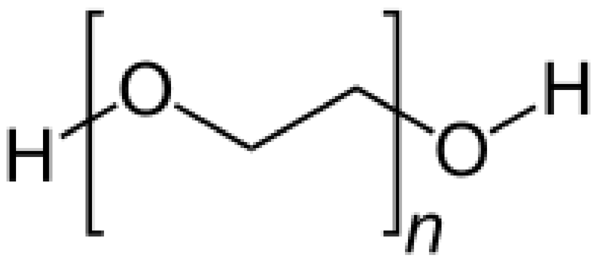
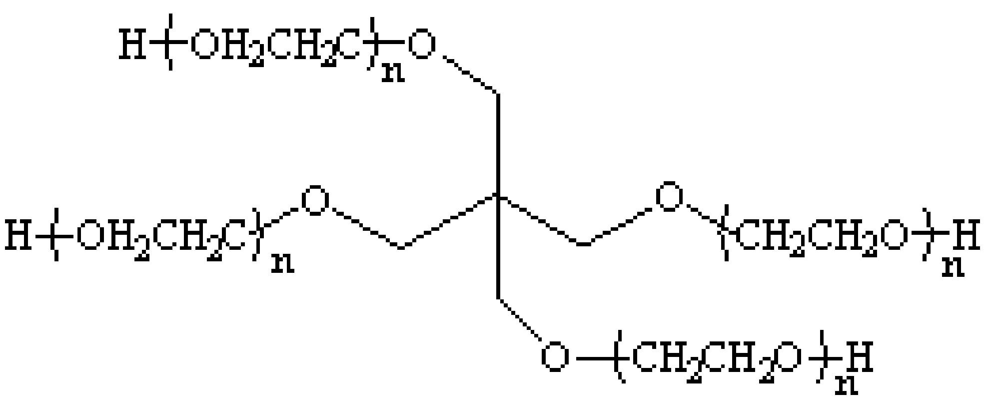
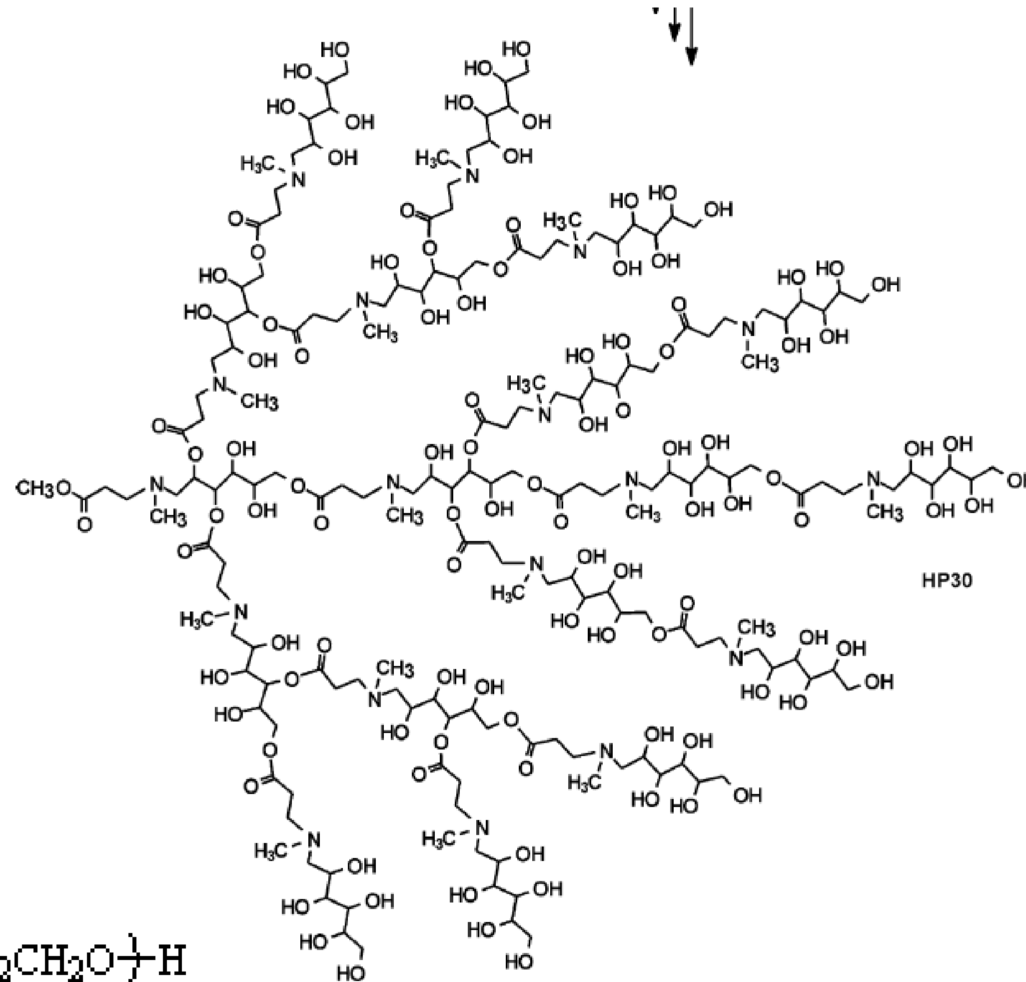


FIG. 2. Polymer arrangements. (From F. Rodriguez, *Principles of Polymer Systems*, Hemisphere Publ., 1982, p. 21, with permission.)

Linear, multi-arm, & Hyperbranched PEG





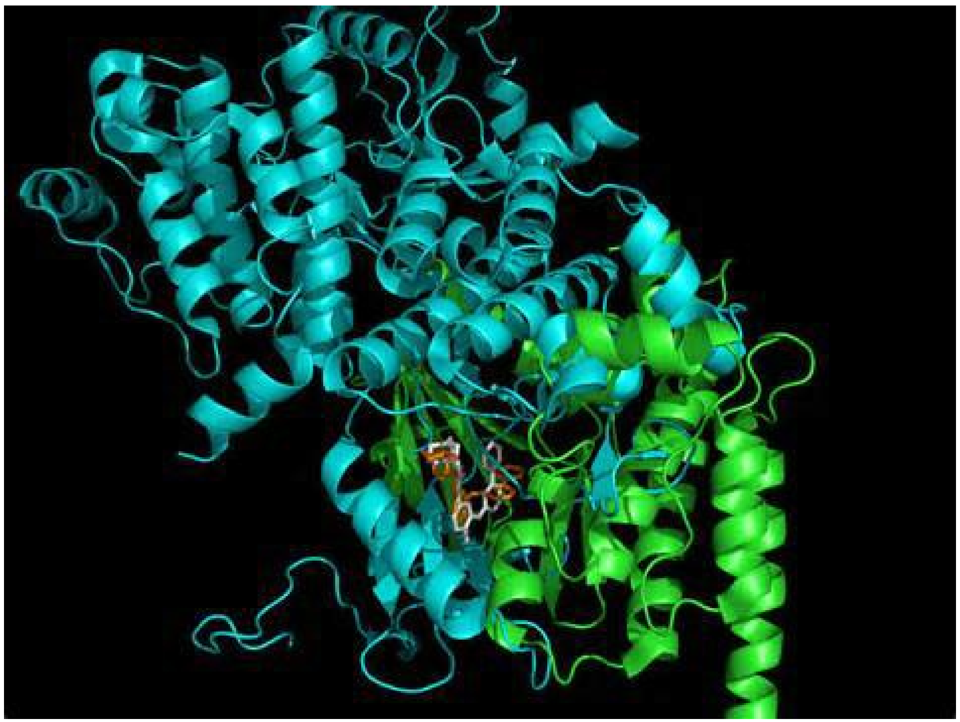


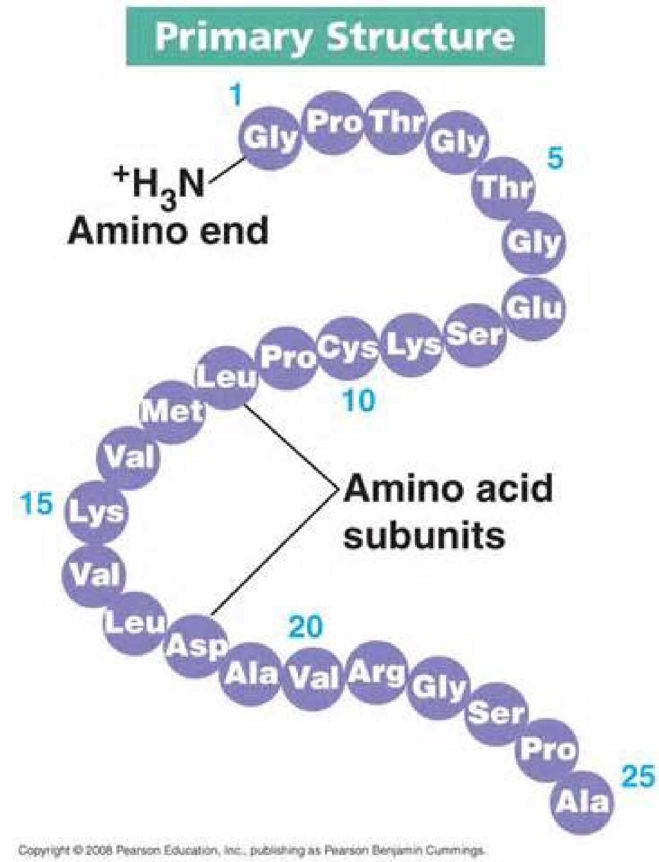
Scheme 18.

Other Linear Polymers

- Proteins are linear polymers that consist of all levo-isomers of amino acids.

左旋异构体





Dextrorotation and levorotation

能使平面偏振光向左旋转的光学异构体。通常用(-)表示左旋。

大自然里充满了各种各样有趣的现象,**“左旋”和“右旋”**，就是这些现象之一。

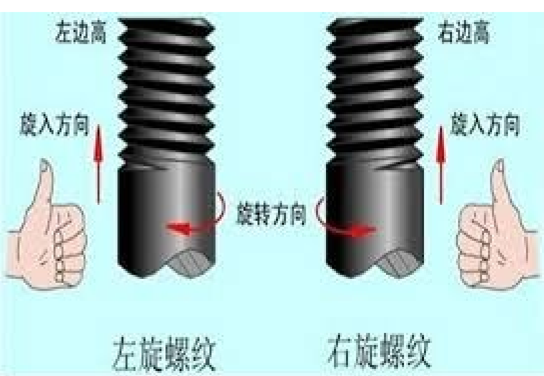
比如，牵牛花的藤 (**vine of the morning glory**) 总是向右旋转着往上长的，这种右旋，“逆时针方向”。

我们吃的糖，无论甘蔗汁制的还是甜菜汁 (**sugar cane juice or beet juice**) 制的，它们分子的几何形状都是右旋的。近年来人工合成了左旋糖，但是说也奇怪，这种糖只有甜味，却不产生热量，原来我们身体里的代谢酶，只接受存在于自然界里的右旋糖。

不过左旋糖也有它特殊的作用，正因为我们机体不能消化吸收它，可它又具有甜味，所以在制作低热值的甜食，或应用于像糖尿病之类病人时，就大有用处了。

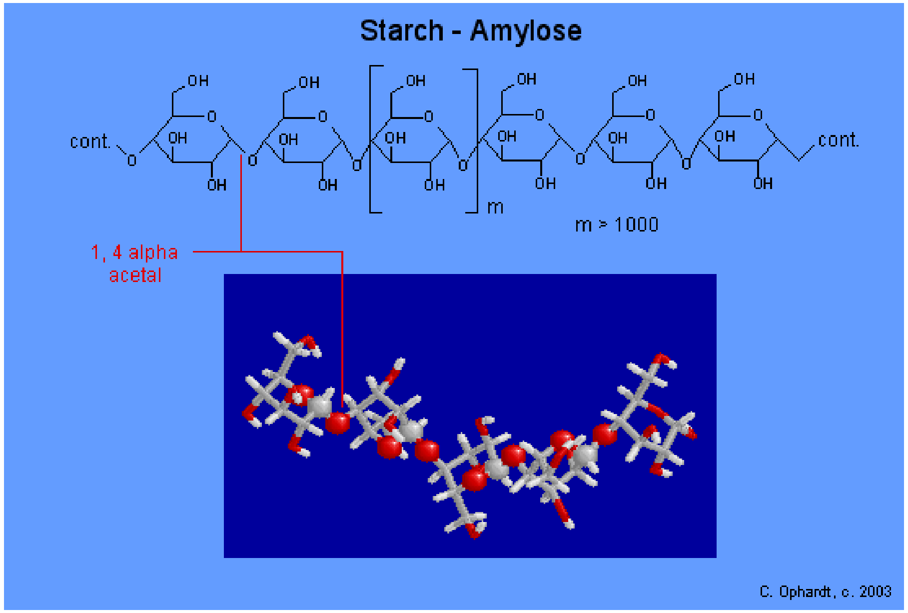
在人体内，一切氨基酸分子都是左旋的，而淀粉的分子却都是右旋的，

传递生物遗传信息的脱氧核糖核酸，它那巨大的分子有着盘梯一般的双螺旋形状，这种螺旋从底部到顶端，一路都呈右旋。科学家最近发现了一种呈锯齿形的核酸，它却是左旋的

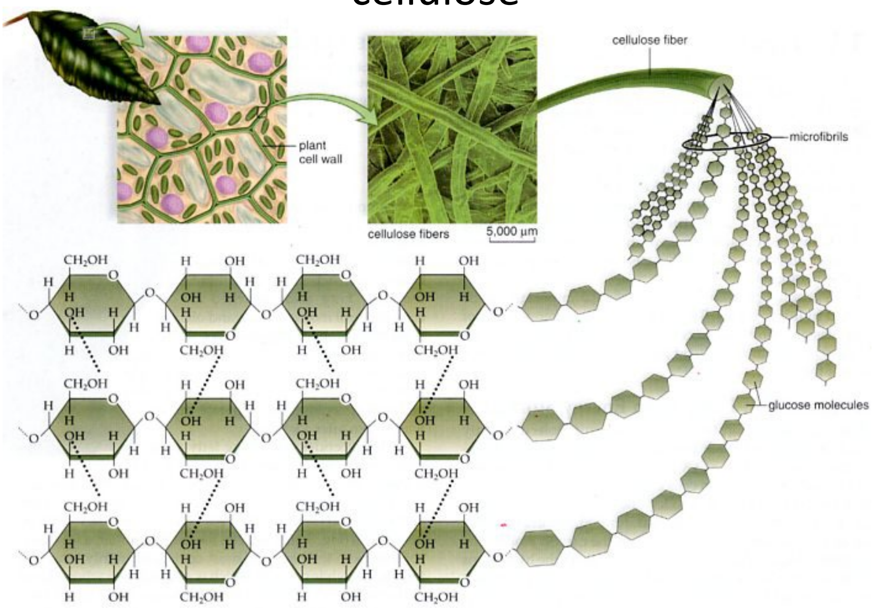


Other Linear Polymers

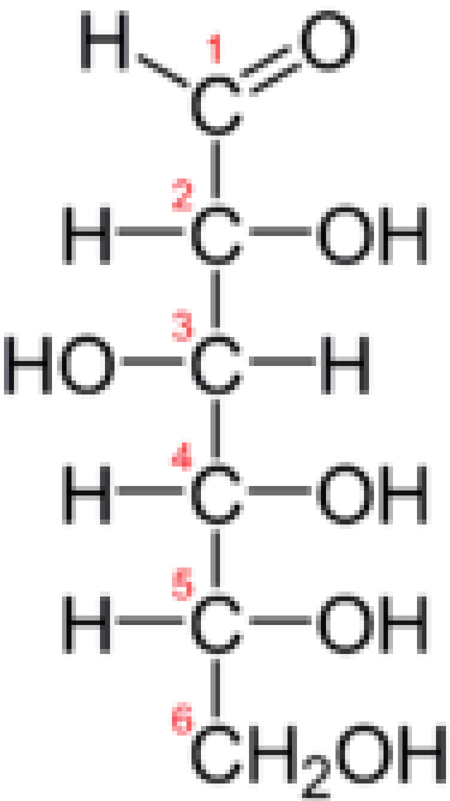
- In contrast, the building blocks of starch and cellulose are d-glucose and are joined by both condensation through <u>both alpha and beta acetal groups</u>.



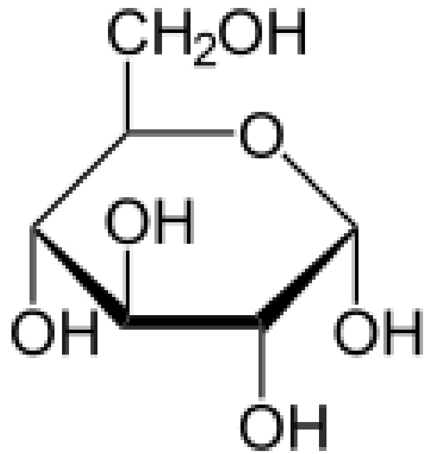
cellulose



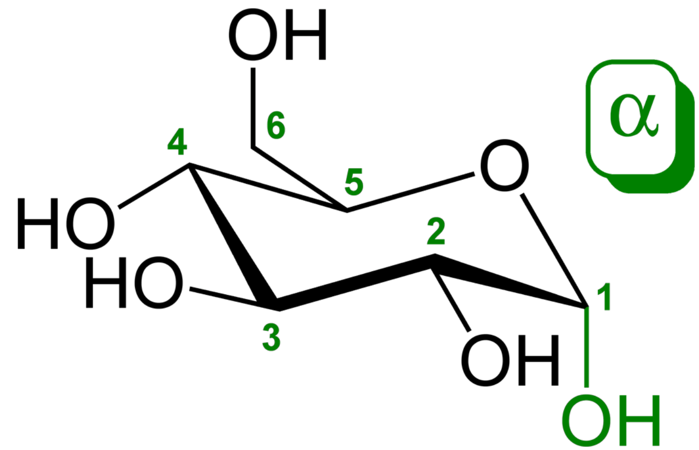
Glucose



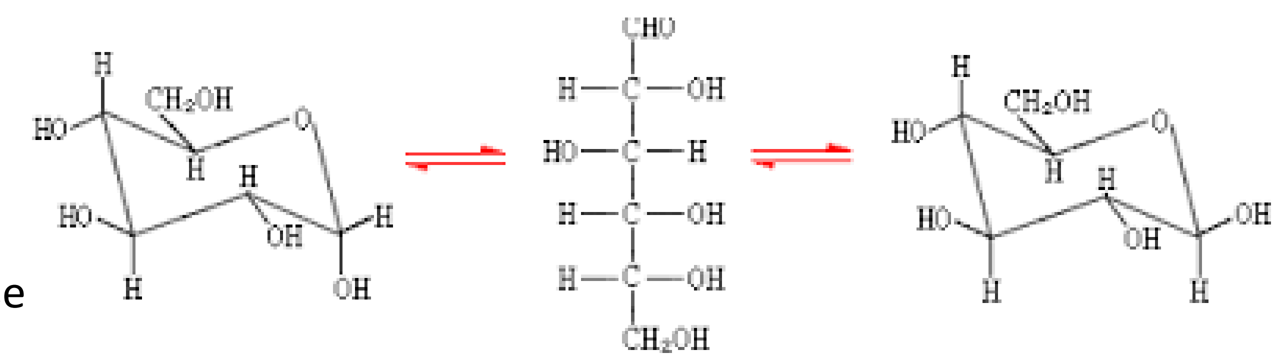
Fischer projection of D-glucose



Haworth projection of α-D-glucopyranose

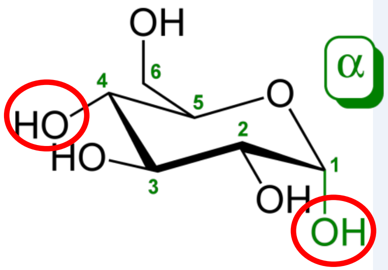


α-D-glucopyranose (chair form).

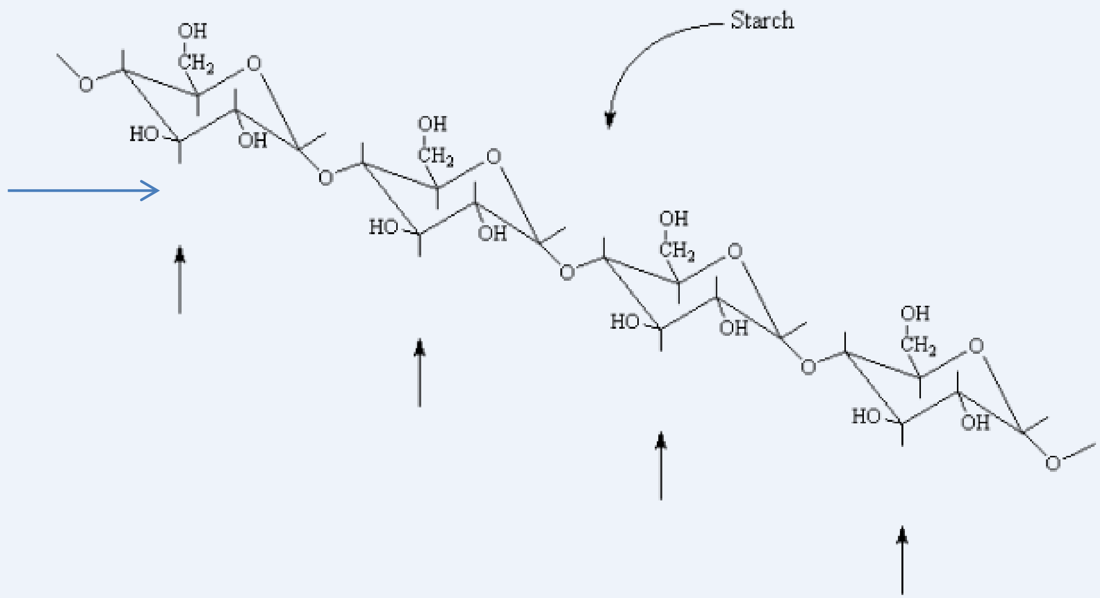


α 型 36%　　D-(+)-葡萄糖≈ 0.1%　　β 型 64%

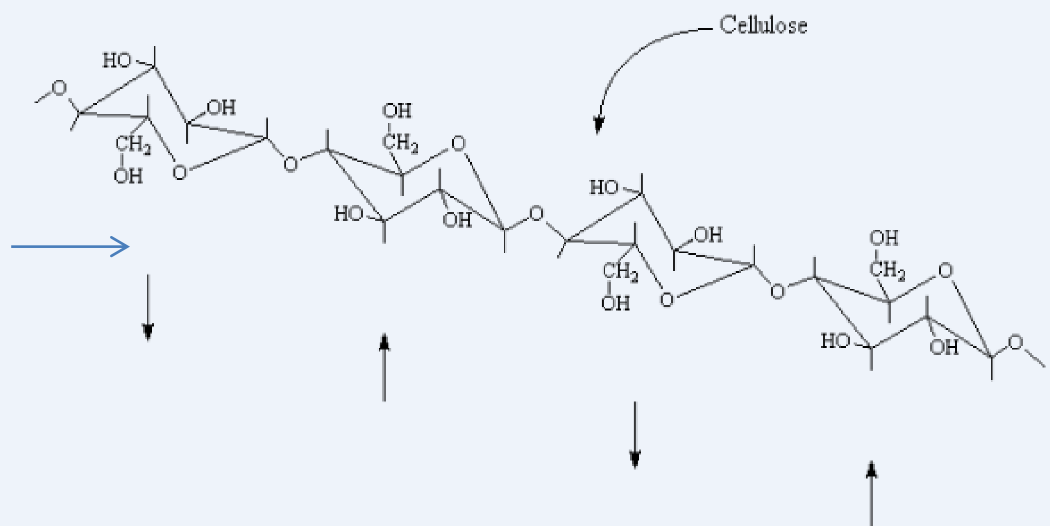
Starch and Cellulose

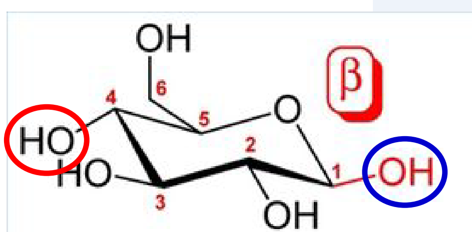


alpha-glucose



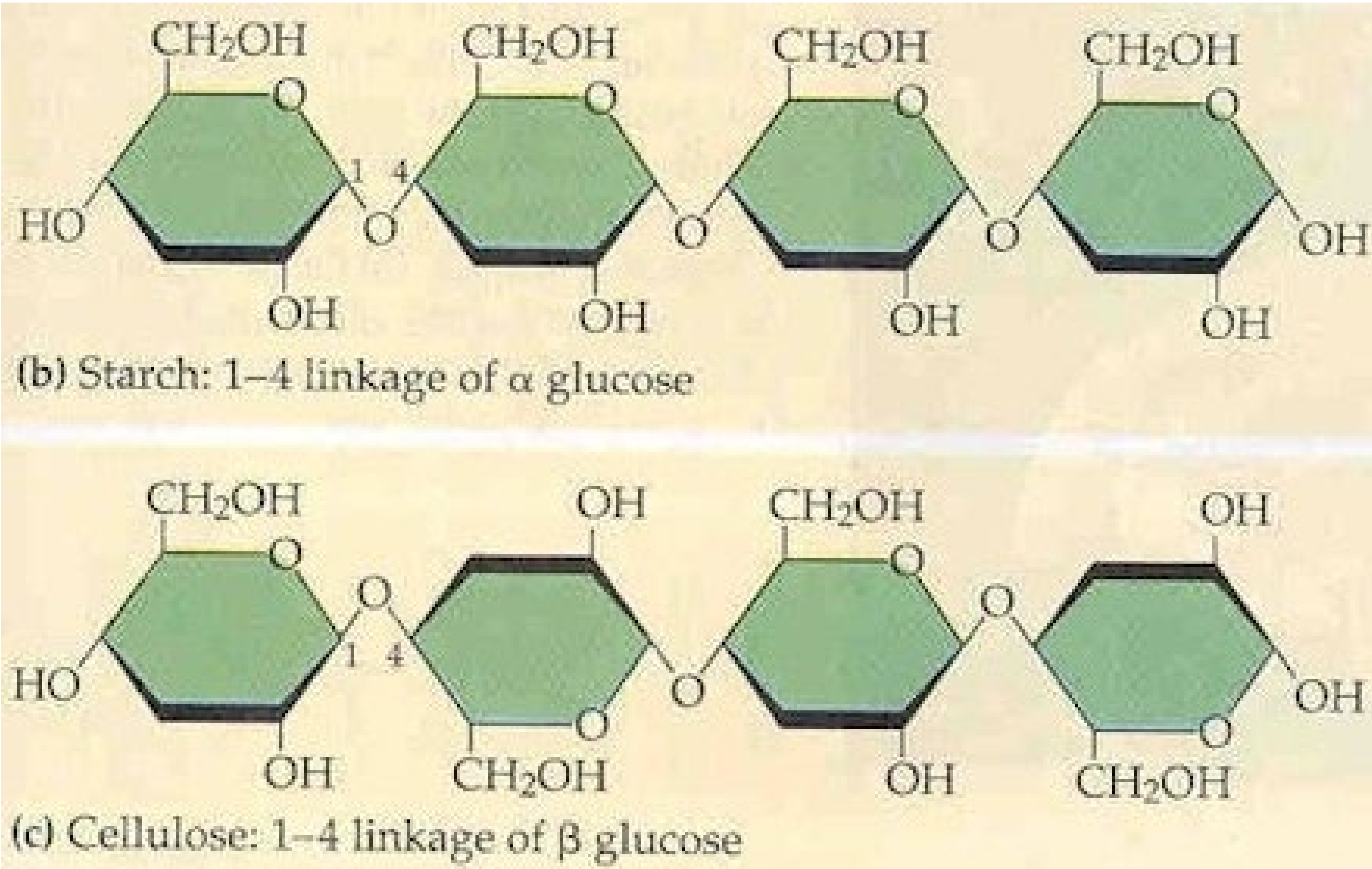
In starch, all the glucose repeat units are oriented in the same direction. But in cellulose, every other repeat unit is rotated 180 degrees around the axis of the backbone, relative to the previous repeat unit.





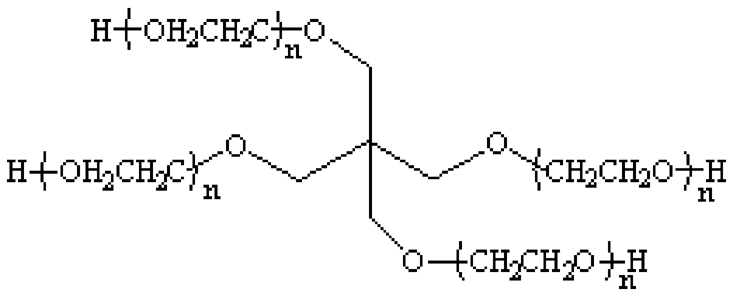
Beta-glucose

Starch and Cellulose



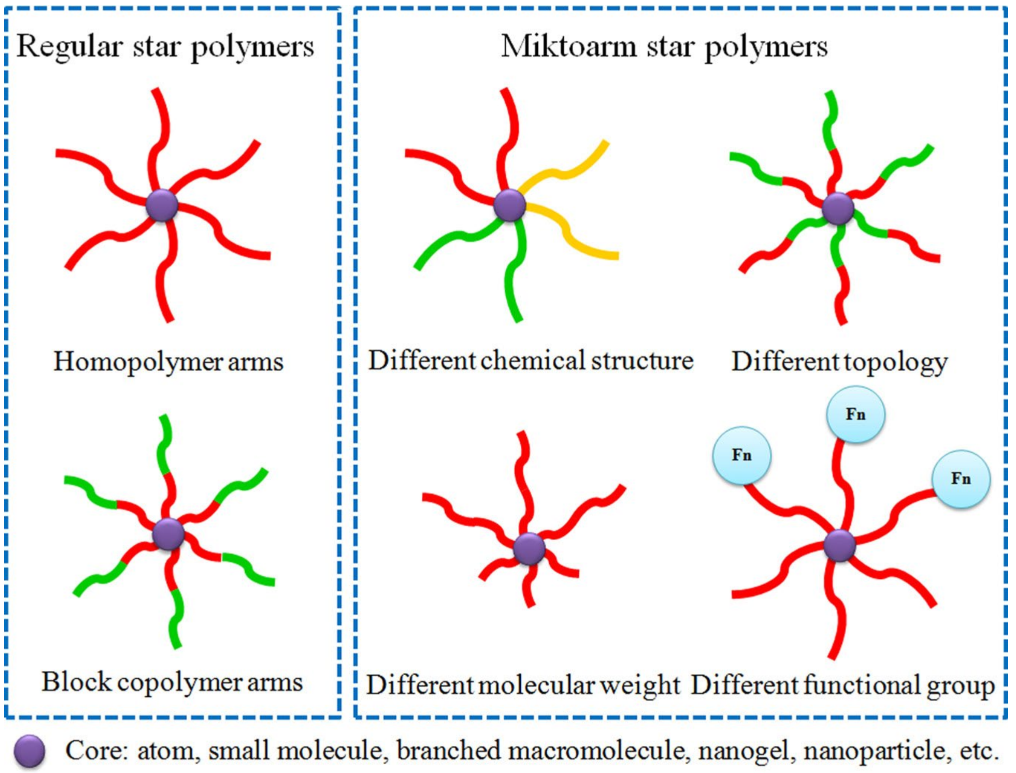
Star Polymers

- Sometimes the ends of several polymer chains are joined together at a common center.



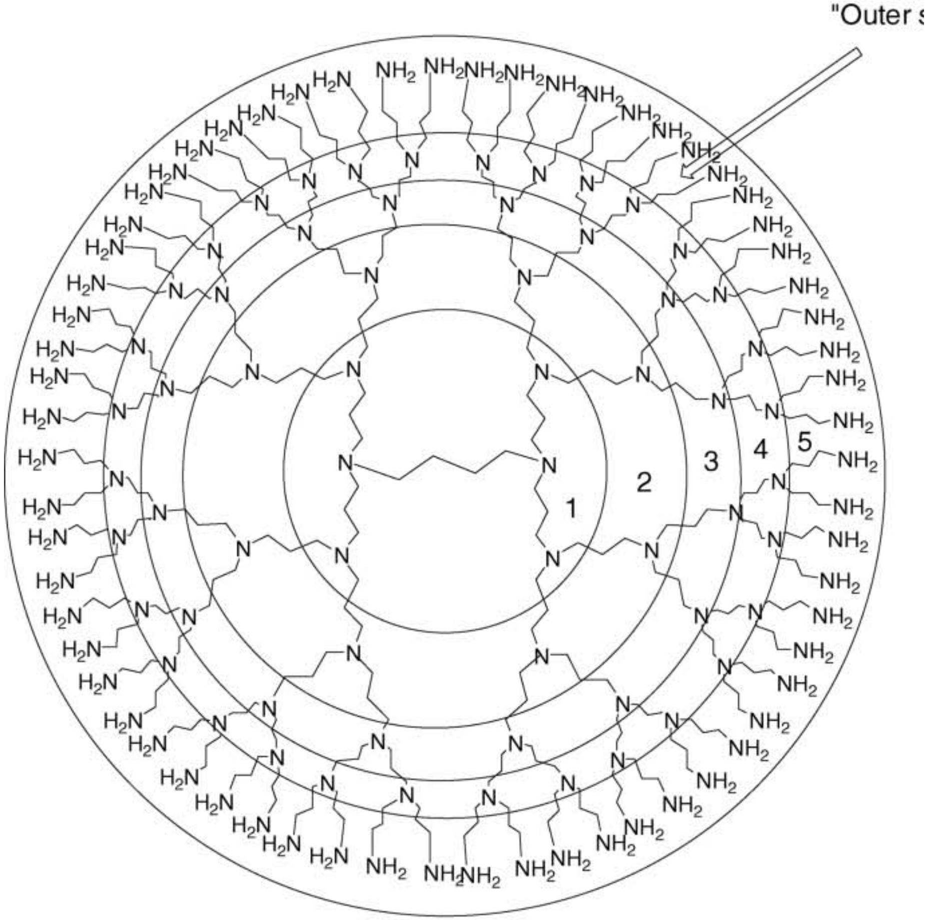
- Polymers like this are called *star polymers*.

- They're often used as additives or as coating materials.



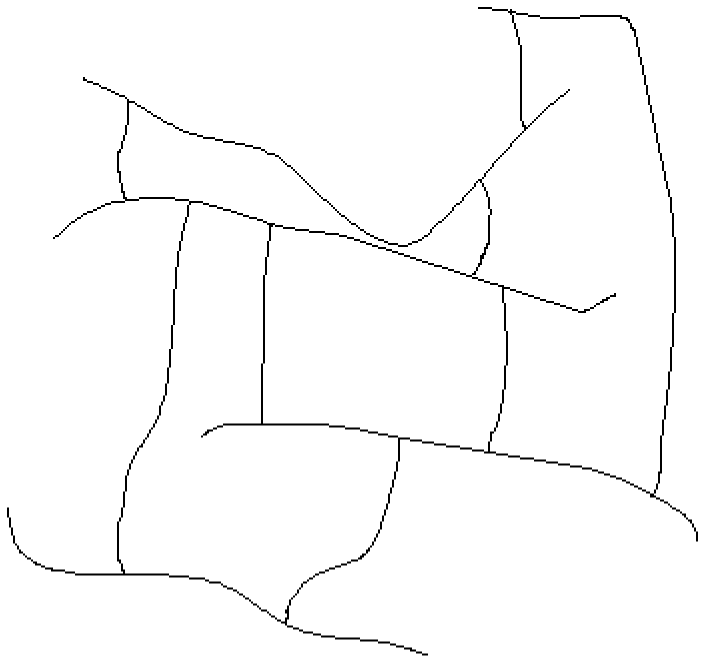
Dendrimer

- Sometimes there is no backbone chain at all.
- Sometimes a polymer is built in such a way that branches just keep growing out of branches and more branches grow out of those branches.
- These are called *dendrimers*, from the ancient Greek word for "tree".



Cross-linked Polymers

- Sometimes, *both* ends of the branch chains are attached to the backbone chains of separate polymer molecules.
- If enough branch chains are attached to two polymer molecules, it can happen that *all* of the polymer backbone chains in a sample will be attached to each other in a **giant 3-D network.**
- This is what happens in **certain hydrogels, polyelectrolytes, rubber, silicone and certain polyurethanes.**



a crosslinked polymer

What makes One Polymer Different from Another?

- Strength of intermolecular forces and their sum over long polymer chains.

- Molecular weight and entanglement, which slow down motion of polymers.

- Crystallinity.

- Crosslinking.

All these properties determine the diverse states of macromolecular aggregation that polymers show.

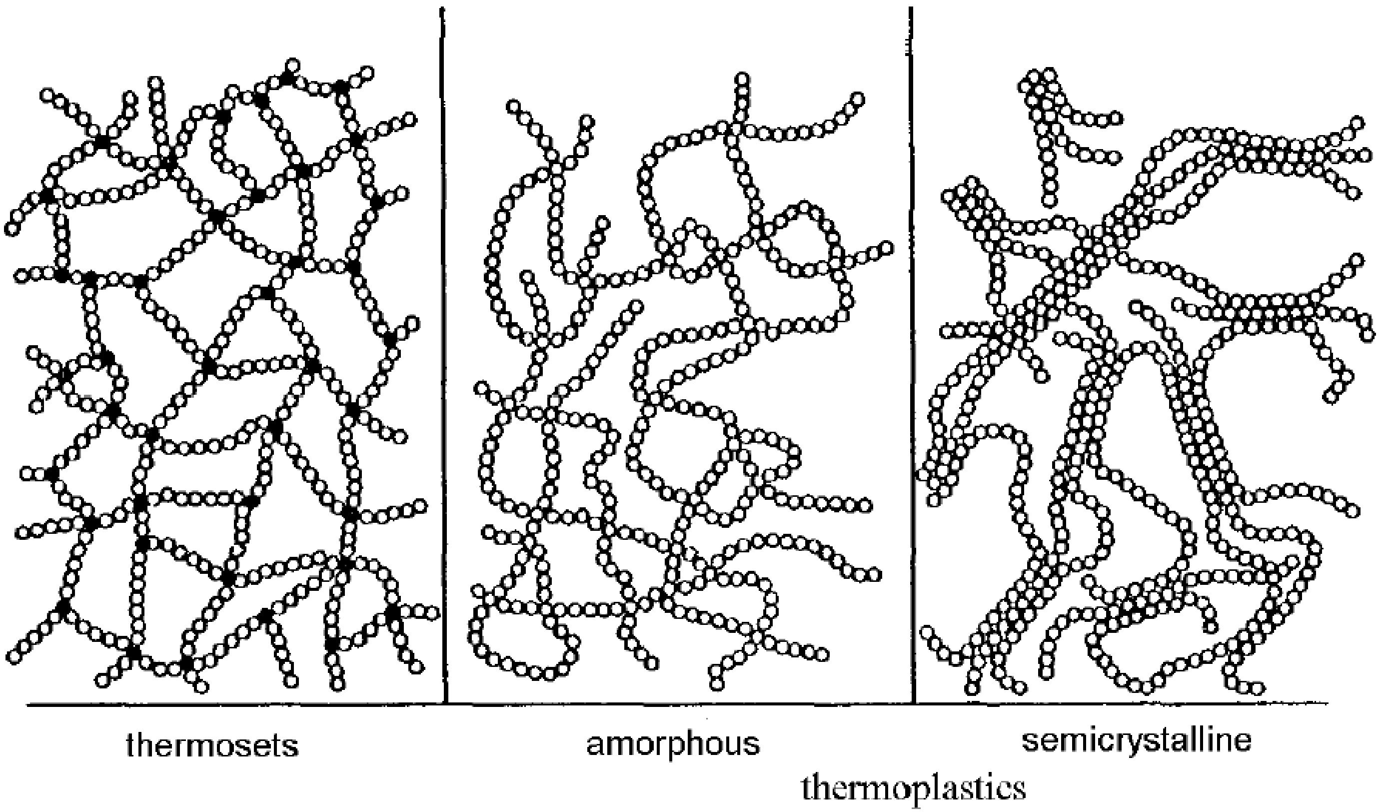
Types of Polymers

- Thermosets
- Thermoplastics

Classification based on Processing

- Elastomers – Classification based on mechanical properties
- Hydrogels- Classification based on chemical properties
- Polyelectrolytes-Classification based on chemical properties
- Natural-Classification based on origin
- Biodegradable-Classification based on biostability

Schematic sketch of thermosets and thermoplastics. The latter can be amorphous or a structure similar to thermosets but a lower crosslink density.



Binding and Structures of Polymers

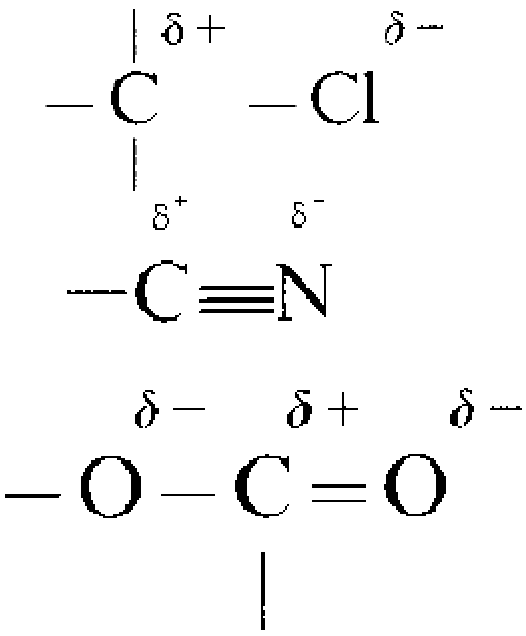
Interchain bonding: covalent

Intermolecular binding

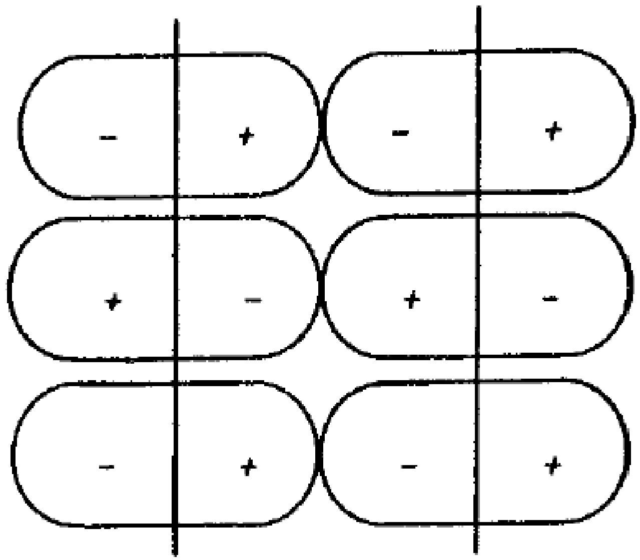
- permanent dipole (polar groups)
- induction forces: induced dipole
- hydrogen bonds
- repulsive forces (Pauli principle)
- Van der Waals interaction

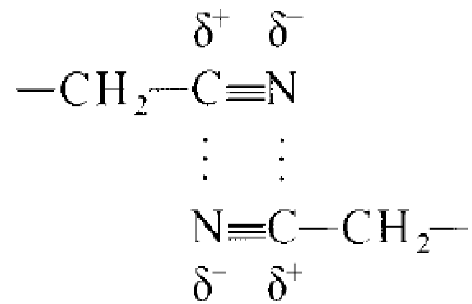
Intermolecular Interactions

- Forces between permanent dipoles
- Different electronegativity of partners
- permanent dipole moment
- Examples of “polar groups“: e.g., in PVC



Dipole forces in a polymer.





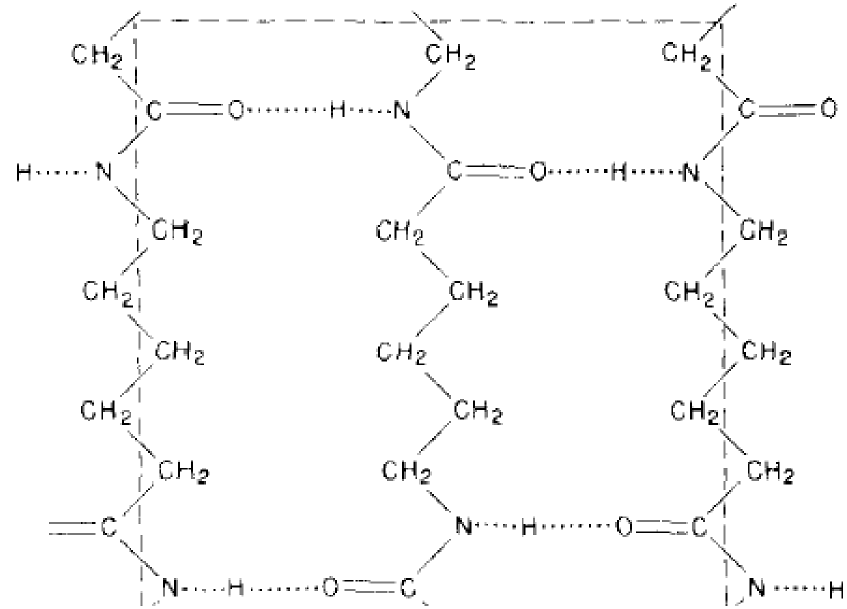
Effect of Polar Groups:

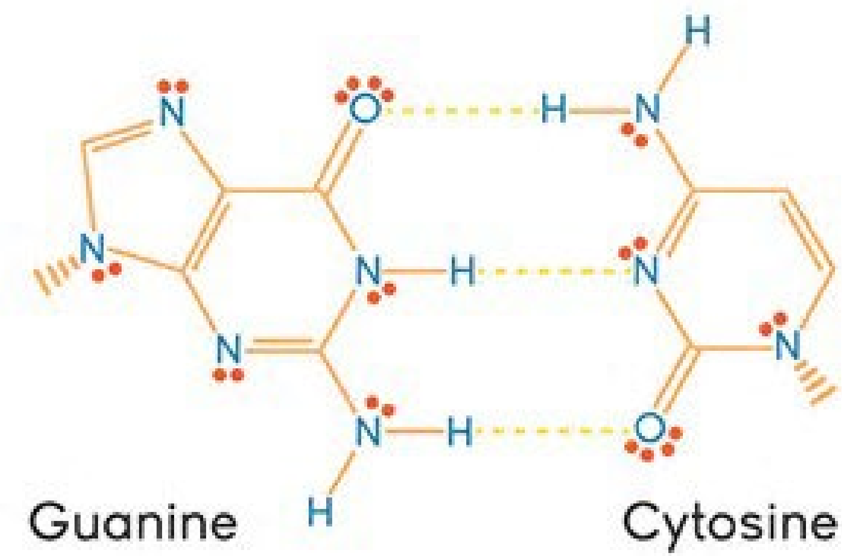
- lower solubility (except in strongly polar solvents)
- higher softening temperature (glass temperature *T*g).

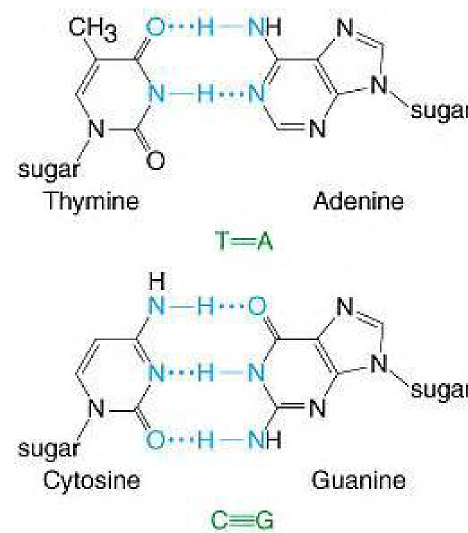
Hydrogen bonds

only for F, O, N as strongly electronegative partner

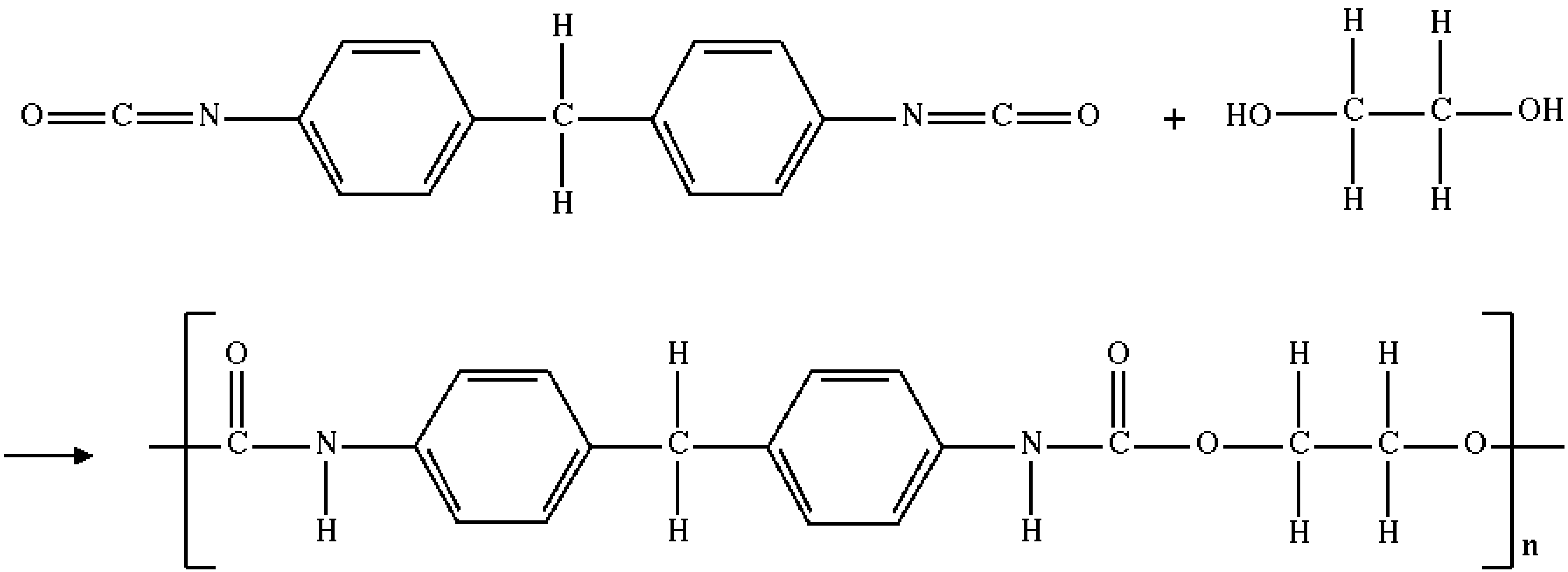
- Illustration of hydrogen bonds in polyamid 6 (PA6)
- Particularly strong in polyamides and polyurethanes

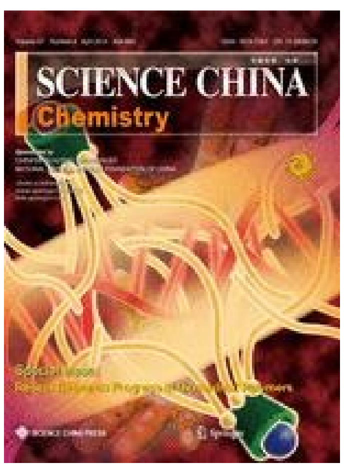


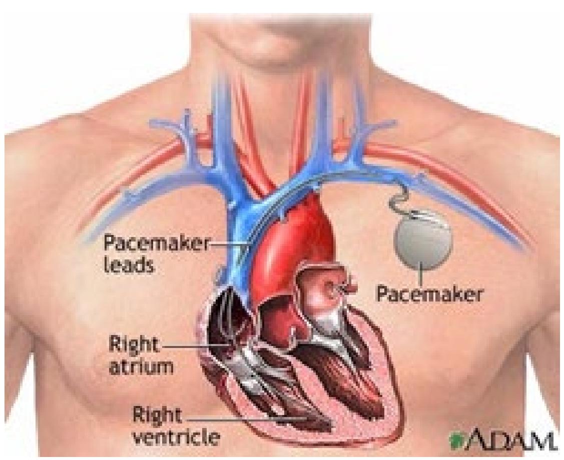




Polyurethanes

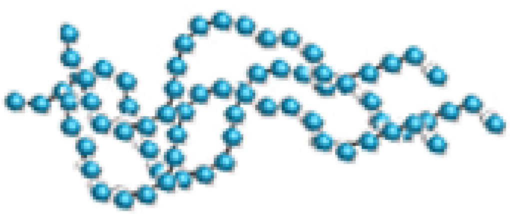


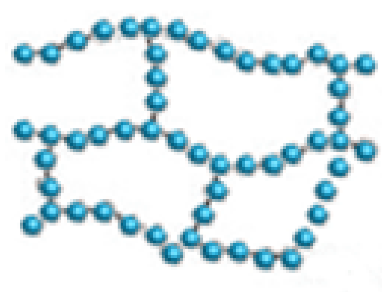


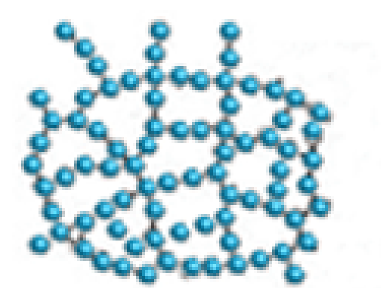


Thermoplastics

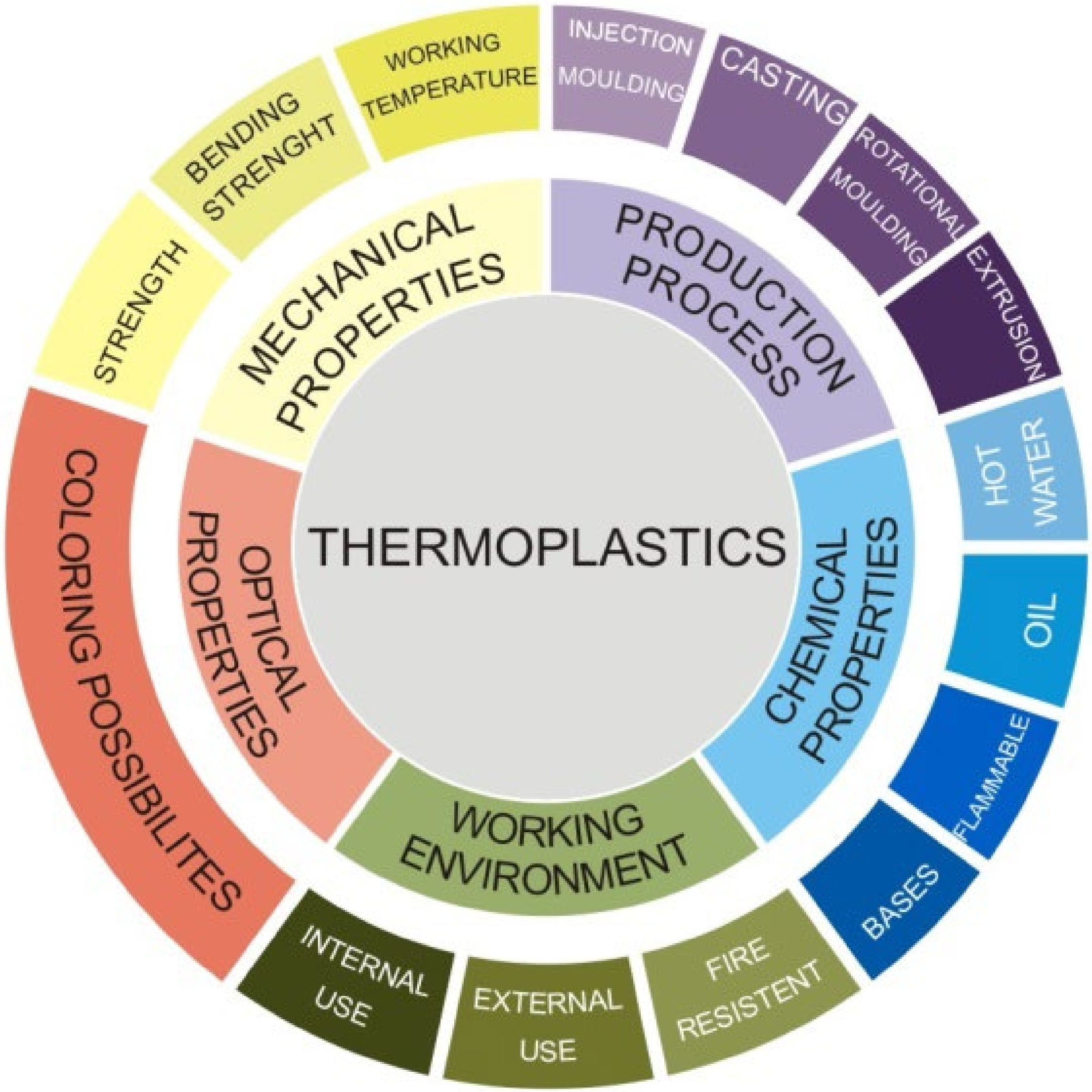
- Thermoplastic polymers are defined as materials that soften, melt, and flow when heat is applied; the adhesives solidify when cooled.
- Majority of familiar plastics
- Can be reprocessed


Thermoplastic


Elastomer


Thermoset

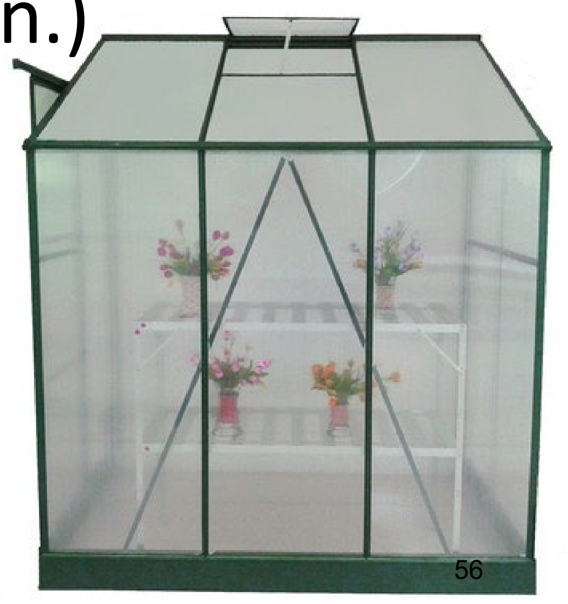
Thermoplastics



Thermoplastics

Amorphous

- Random structure
- Good clarity
- Broad melt temperature
- Low mold shrinkage (<0.005 in./in.)
- Acrylic, polycarbonate, PETG, polystyrene, PVC, TPU,

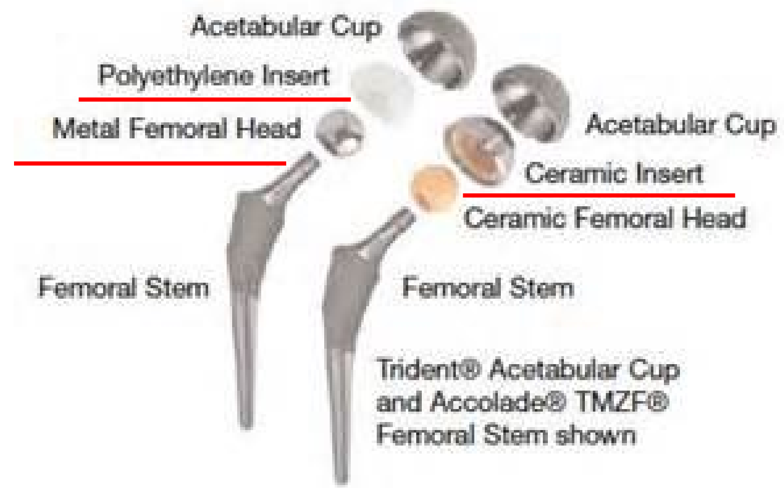


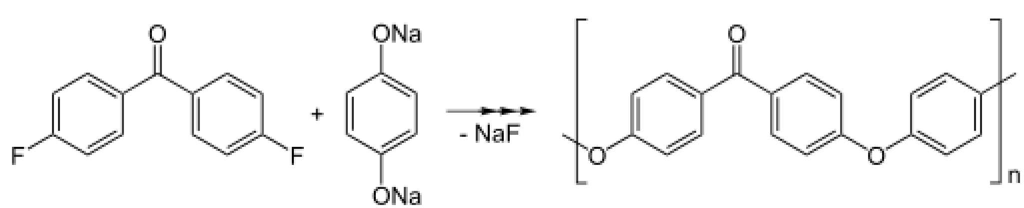
$\alpha = (L0 - L) / L0$
Where, L0: the cavity dimensions (mm) L: Dimensions (in mm) of the molded product at room temperature (usually 20°C).

Thermoplastics

Semi-crystalline

- Linear alignment of chains
- Harder, less flexible
- Unique melting point
- High mold shrinkage (>0.01 in./in.)
- Polyethylene (LDPE / MDPE /HDPE), polypropylene, PTFE, Polyamide, PEEK, TPU (Thermoplastic polyurethane)





Thermosets

- Not meltable
- Not soluble
- Not swellable
- Processing generally prior to crosslinking

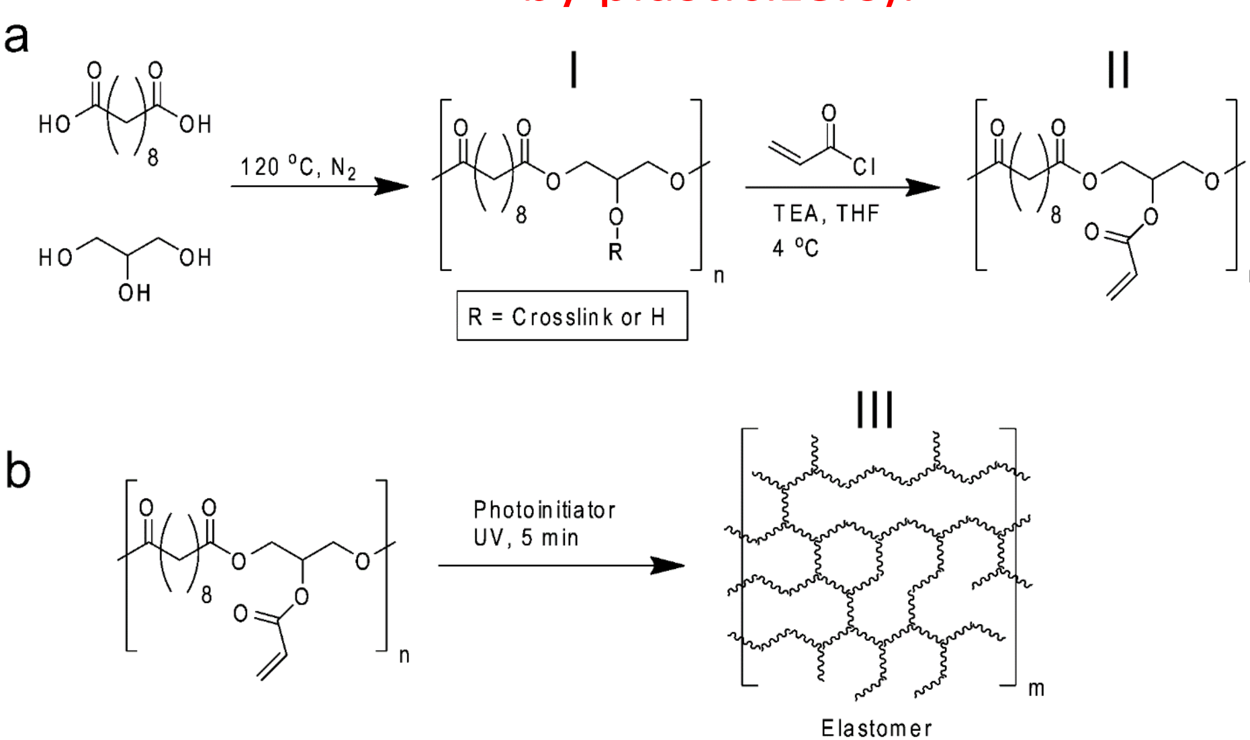
Thermosets

- A thermoset polymer, as the name suggests, becomes set into a given network, normally through the action of a catalyst—heat, radiation, or a combination of these factors—during the process of cross-linking.
- As the name suggests, *cross-linking* is the process of forming cross-links between linear polymer molecules (*curing* is another term commonly used).
- As a result of this process, thermosets become infusible and insoluble.
- **Thermosetting resins (e.g., epoxies, polyesters, and phenolics**) are the basis of many structural adhesives for load-bearing medical applications, as well as for the precision joining of electronic parts.

- Hard, strong, rigid
- Excellent heat resistance
- Cannot be reprocessed
- **Crystalline**: Epoxy, phenolic, polyester
- **Amorphous**: Rubber, silicone, polyurethane

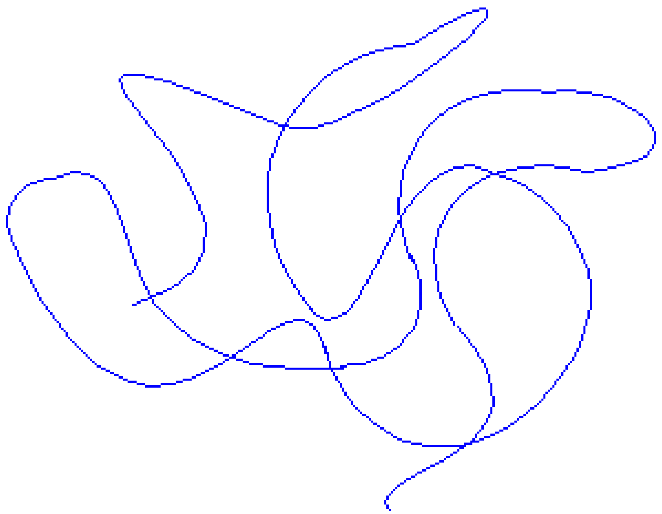
Elastomers

- Not meltable
- Not soluble
- Swellable
- Used at *T* > *Tg* (*Tg* often reduced by plasticizers).

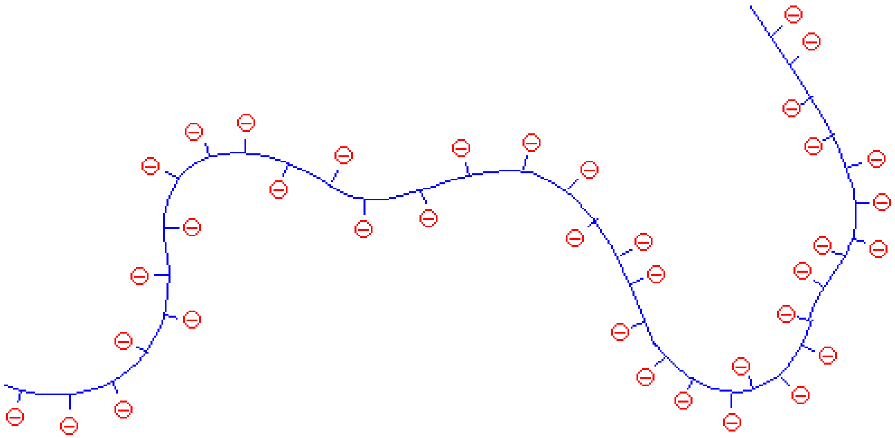


Synthetic biodegradable elastomers for drug delivery and tissue engineering*

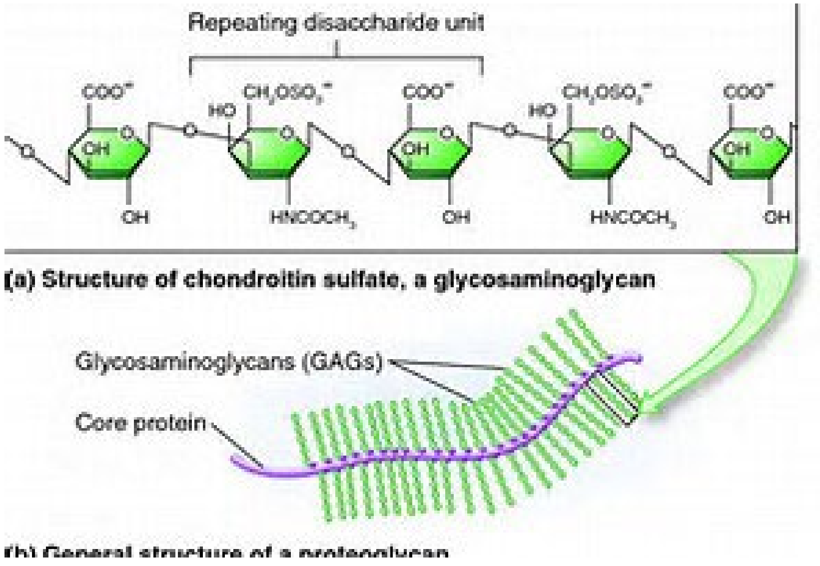
Polyelectrolytes



A polymer molecule tangled in a random coil.



A polyelectrolyte expands because it's like charges repel each other

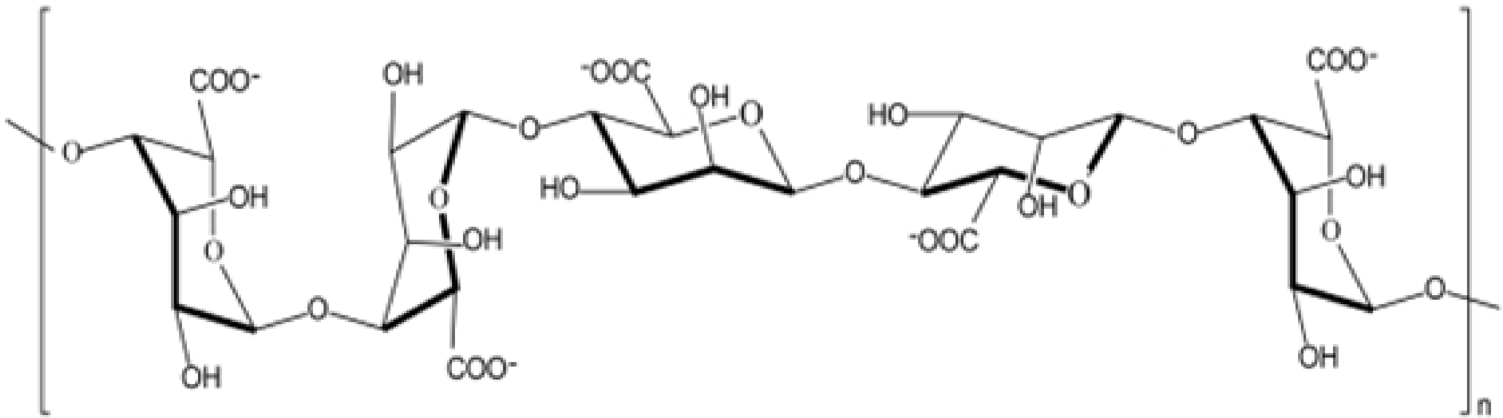


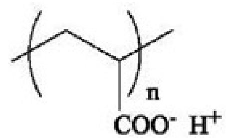
Many biological molecules are polyelectrolytes.

For instance, polypeptides, glycosaminoglycans, and DNA are polyelectrolytes.

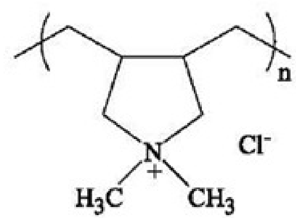
Both natural and synthetic polyelectrolytes are used in a variety of industries.

Polyelectrolytes

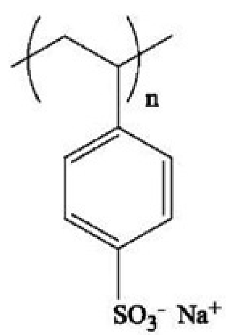




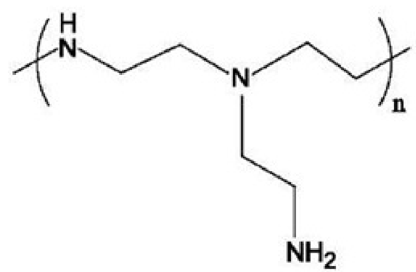
poly(acrylic acid), PAA



poly(dimethyldiallylammonium chloride), PDDA



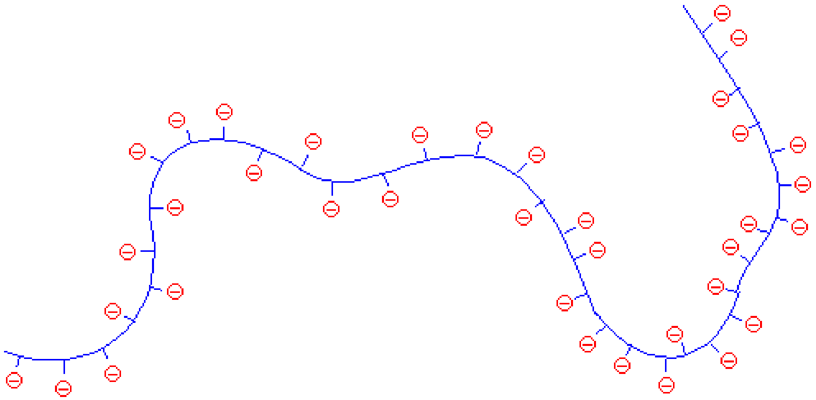
poly(styrenesulfonate) sodium salt, PSS



Branched poly(ethyleneimine), PEI-b

Polyelectrolytes

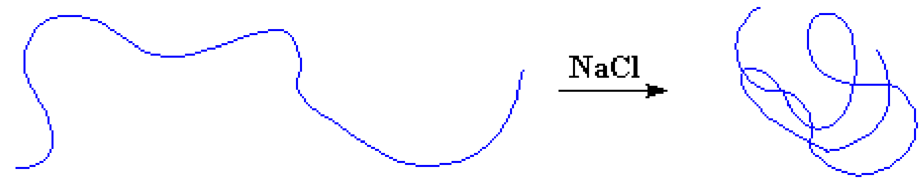
- But when the polymer chain is covered with negative charges (which repel each other), the polymer can't be bunched in on itself. So the chain stretches out, like this.
- This makes the solution (remember we're talking about polyelectrolytes *in solution*) more viscous.
- Think about it.
- When the polyelectrolyte chain stretches out it takes up more space, and is more effective at resisting the flow of the solvent molecules around it.



A polyelectrolyte expands because it's like charges repel each other

Reversibility of the Process

- If one take a solution of a polyelectrolyte in water, and throws in a lot of salt.
- The NaCl will separate into Na+ and Cl- ions.
- In the case of a negatively charged polyelectrolyte like **poly(acrylic acid),** the positively charged Na+ ions will get in between the negative charges on the polymer, and cancel them out in effect. When this happens, the polymer chain collapses back into random coil again.



Salt makes polyelectrolytes in solution collapse into random coils.

DNA提取分为三个基本步骤，每个步骤的具体方法可根据样品种类、影响提取的物质以及后续步骤的不同而有区别。

利用研磨或者超声破碎细胞，并通过加入去污剂以除掉膜脂。

加入蛋白酶，**醋酸盐沉淀**，或者酚／氯仿抽提，以除掉细胞内的蛋白，如与DNA结合的组蛋白。

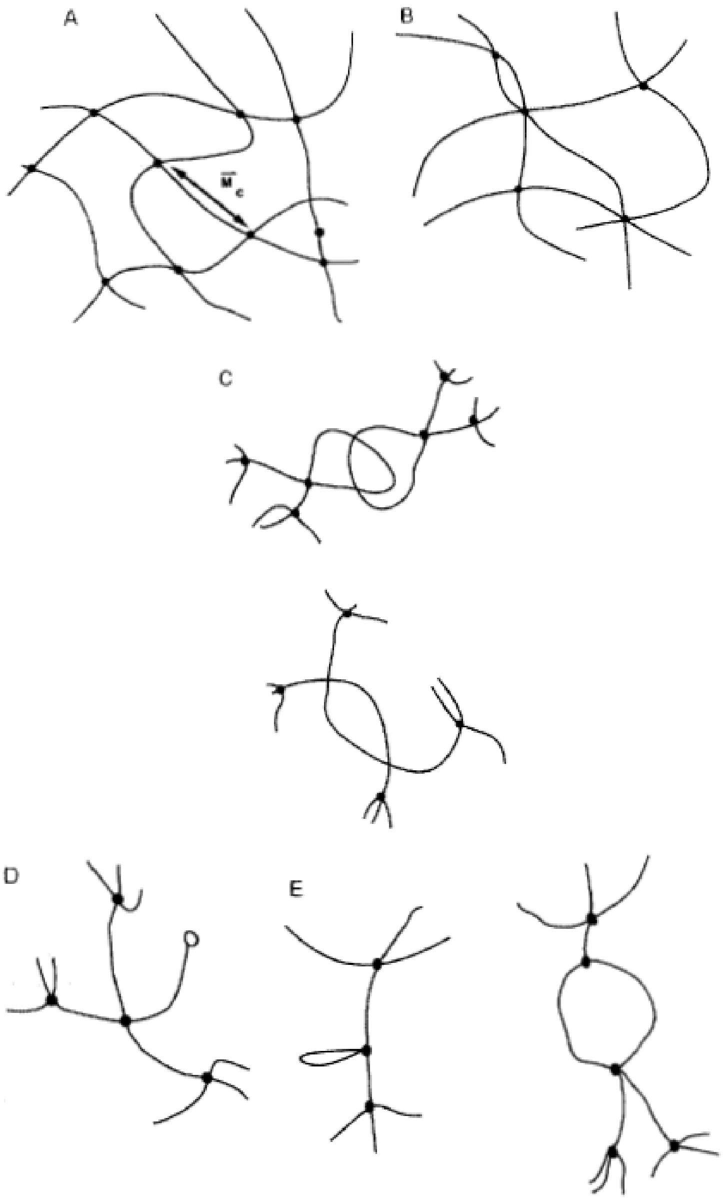
将DNA在冷乙醇或异丙醇中沉淀，因为DNA在醇中不可溶而黏在一起，这一步也能除掉盐分。

Influence of Increasing Molar Mass on Properties

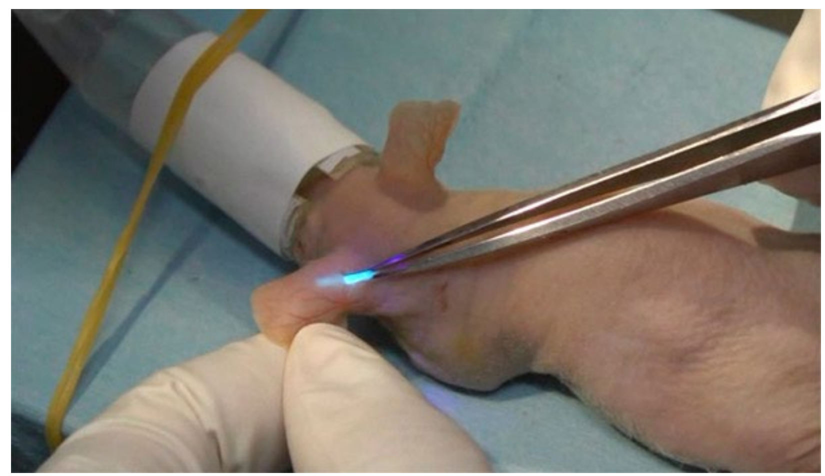
Increasing molar mass leads to			
higher strength	higher impact strength	higher chemical resistivity	reduction in flowability and resistance to melt fracture

Reasons			
higher interchain forces, more entanglements	lower degree of crystallization at higher chain length, more entanglements	higher interchain forces, degradation not so detrimental since high level	more entanglements

Hydrogels



(A) Ideal macromolecular network of a hydrogel. (B) Network with multifunctional junctions. (C) Physical enta
a hydrogel. (D) Unreacted functionality in a hydrogel. (E) Chain loops in a hydrogel.

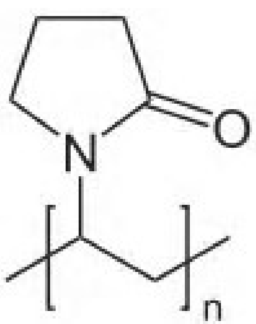


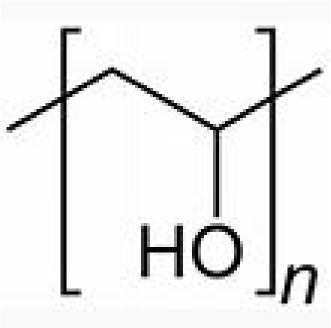
Hydrogels

In general, highly swollen hydrogels are those of cellulose derivatives, poly(vinyl alcohol), poly(N-vinyl 2-pyrrolidone) (PNVP), and poly(ethylene glycol).

Moderately and poorly swollen hydrogels are those of poly(2-hydroxyethyl methacrylate) (PHEMA) and many of its derivatives.

Of course, one may copolymerize a basic hydrophilic monomer with other molecular weight chains through its di- or multifunctional more or less hydrophilic monomer to achieve desired swelling properties





Hydrogels

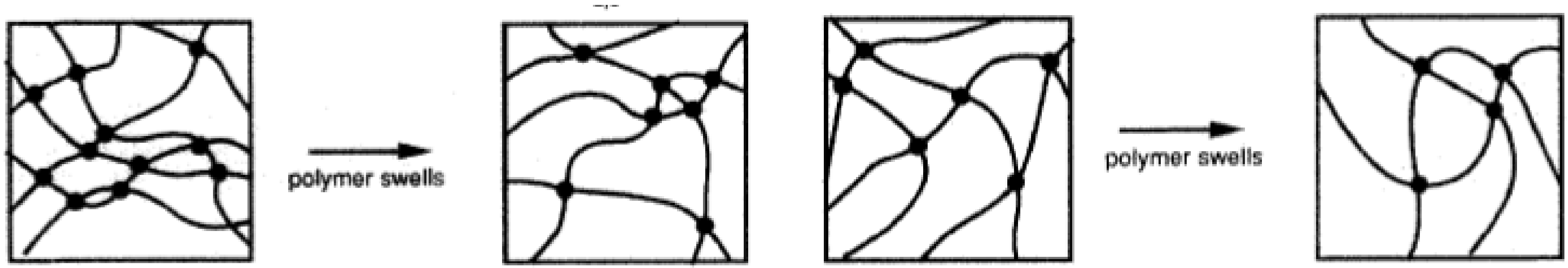
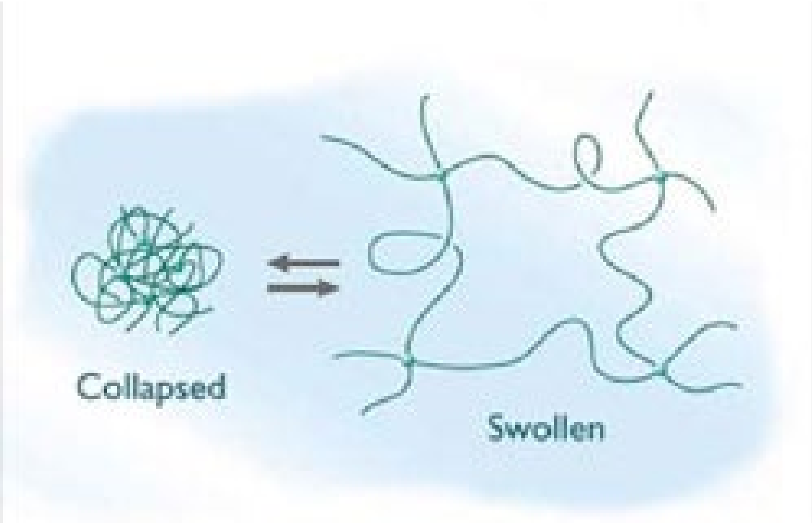
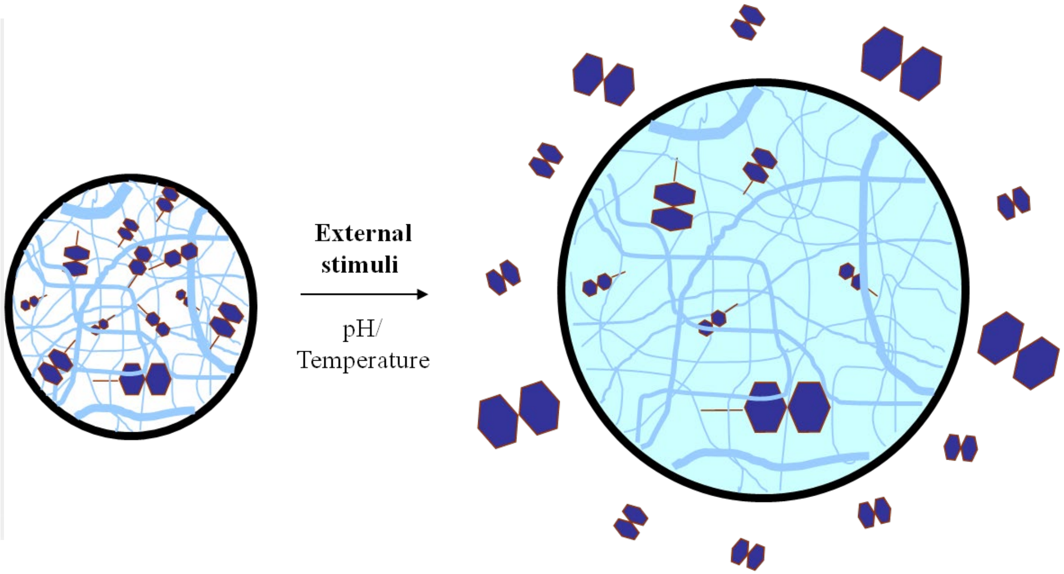


FIG. 2. (A) Swelling of a network prepared by cross-linking in dry state. (B) Swelling of a network prepared by cross-linking in solution.





Hydrogels

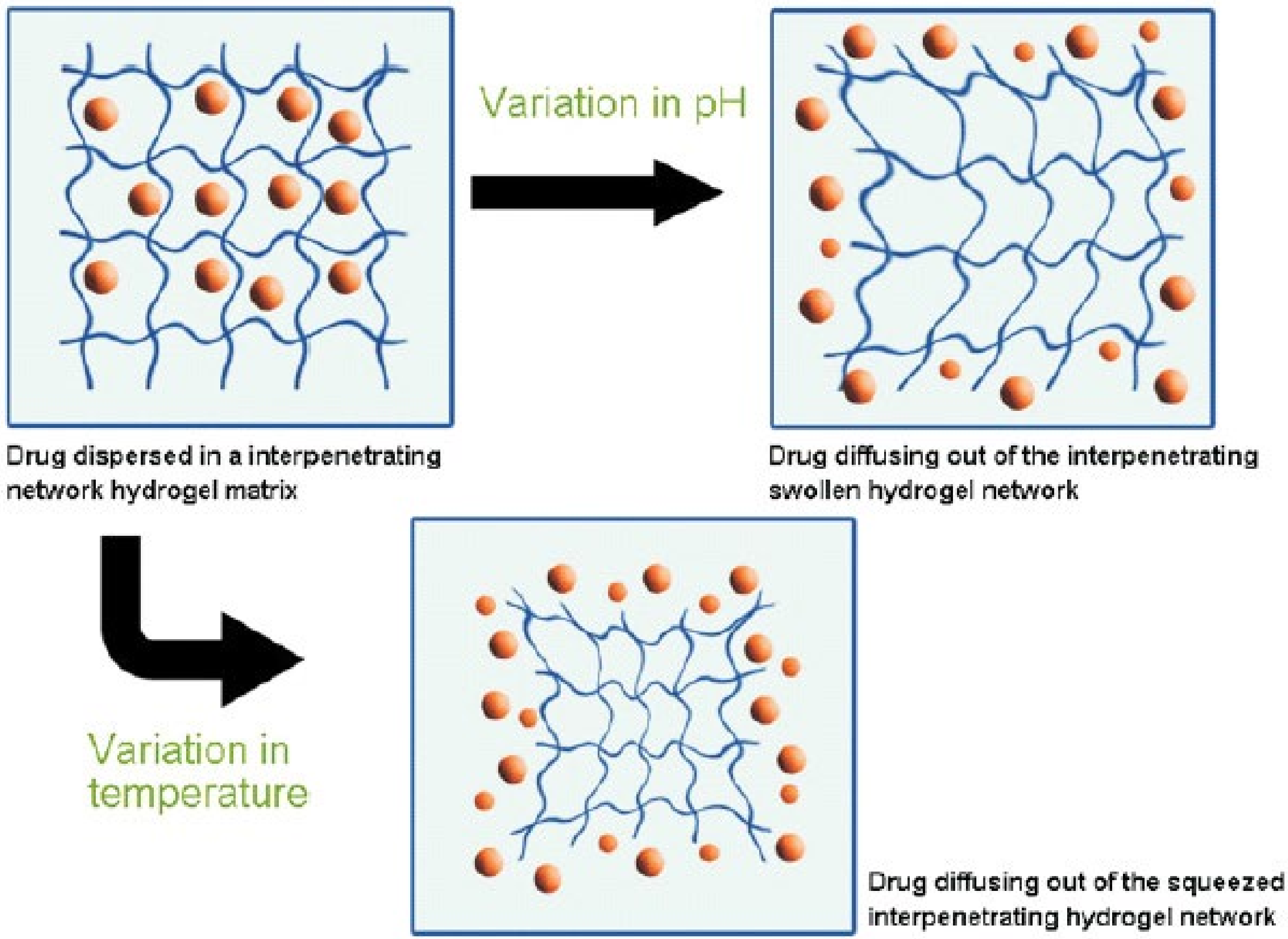
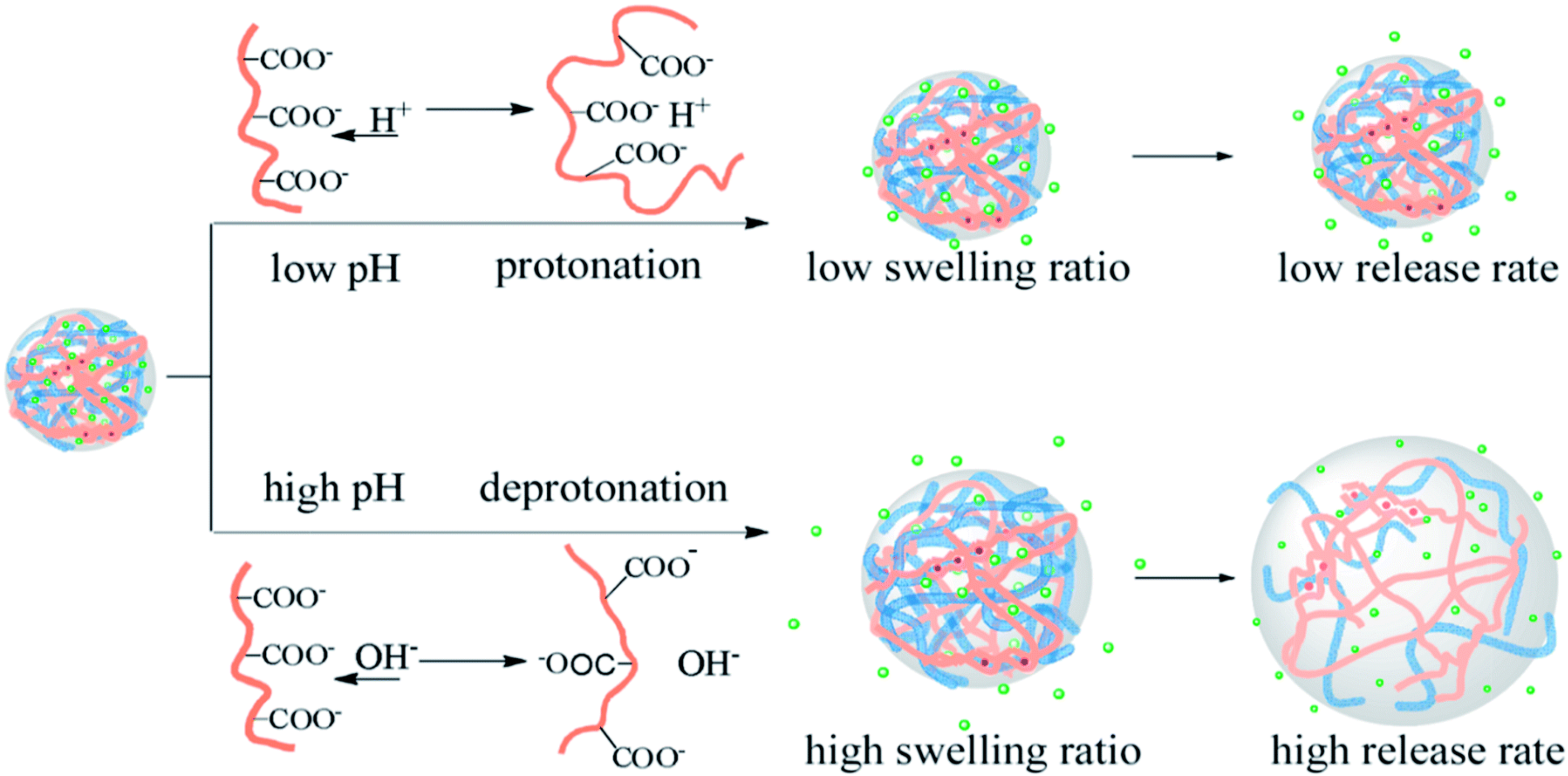


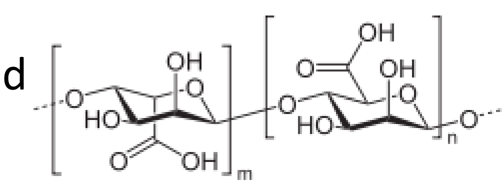
Fig. 1: Swelling and deswelling behaviour of interpenetrating hydrogel network with the variation in temperature and pH

Hydrogels



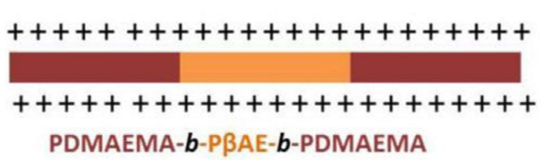
Releasing process for the drug loaded nf-BC/SA hybrid hydrogels under different pH values

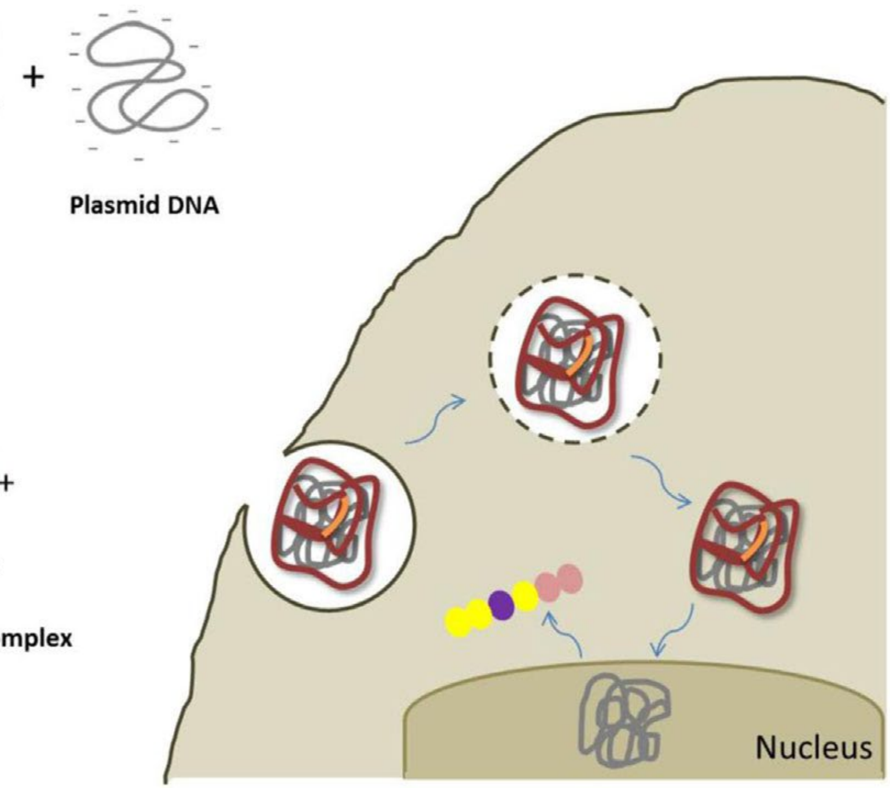
Bacterial cellulose nanofibers (nf-BC) and sodium alginate (SA)



RSC Adv., 2014, 4, 47056

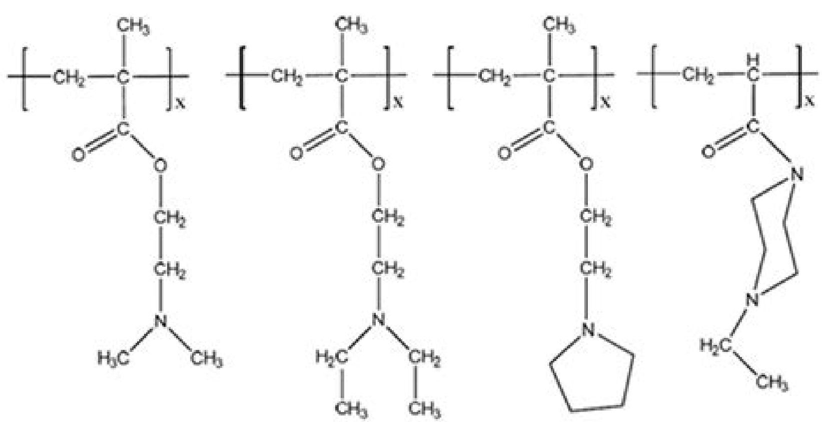
Drug and/or gene delivery



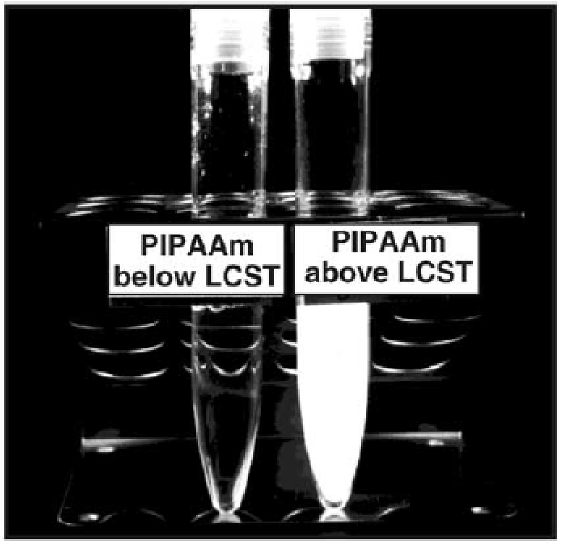


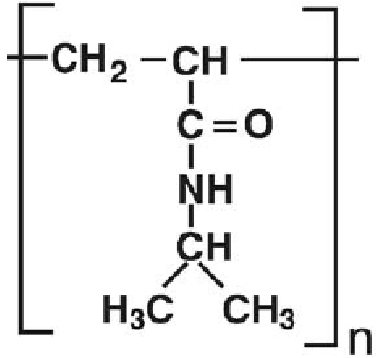
Poly-[2-(dimethylamino)ethyl methacrylate]-block-poly(b-amino ester)-block-poly-[2-(dimethylamino)ethyl methacrylate]:

Macromol. Biosci. 2015, 15, 215–228

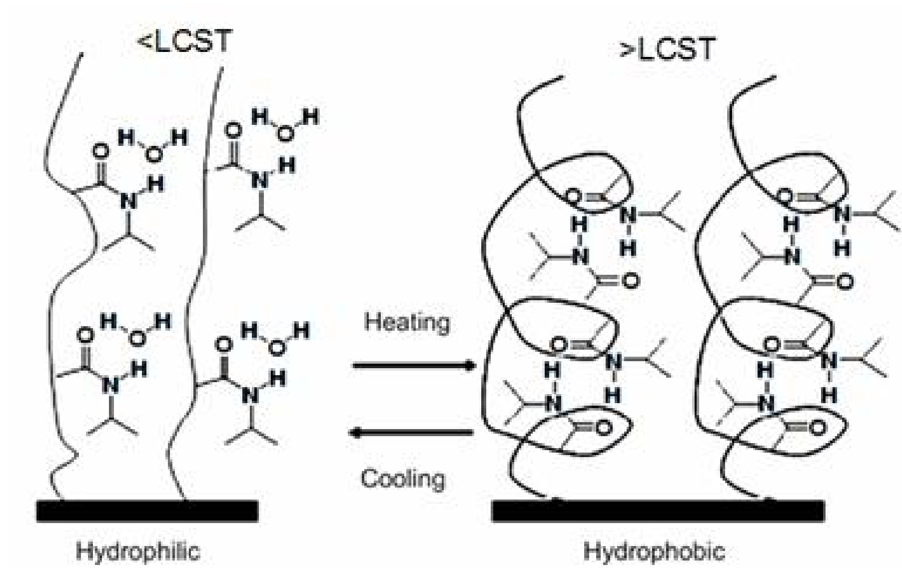


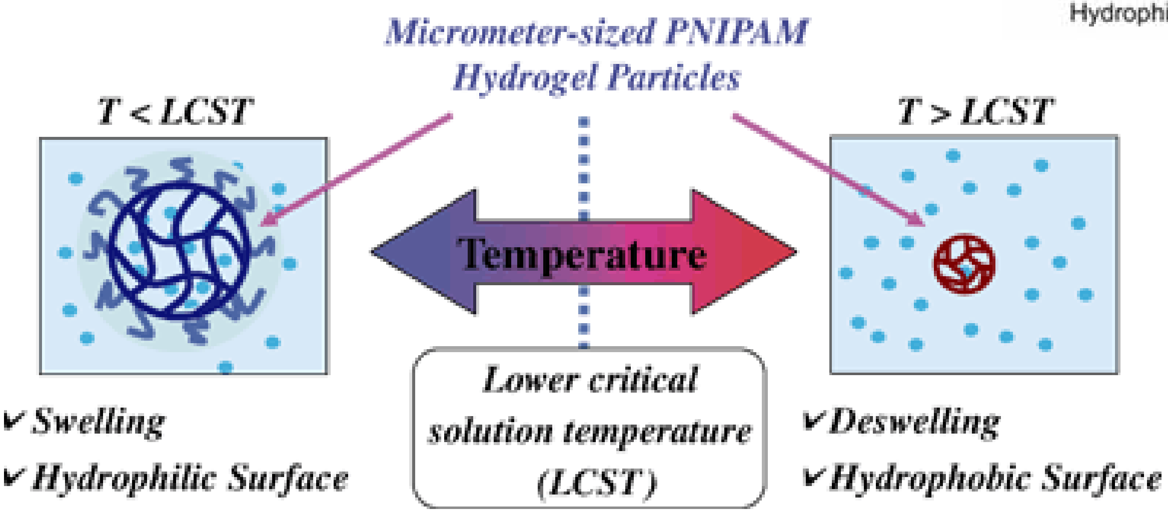
Hydrogels





LCST: 32℃



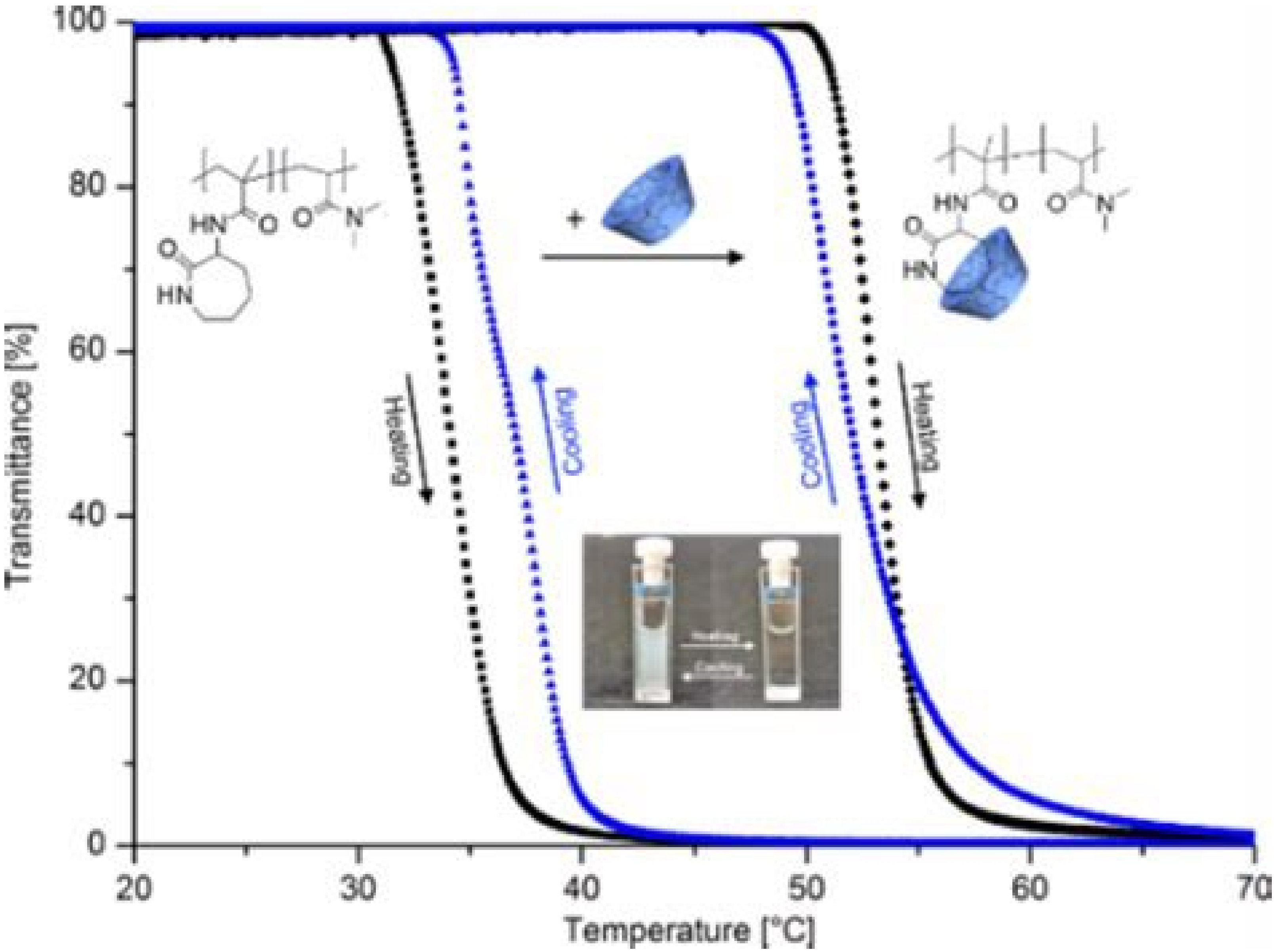


Poly(*N*-isopropylacrylamide): (PNIPAAm):

Low critical solution temperature (LCST)

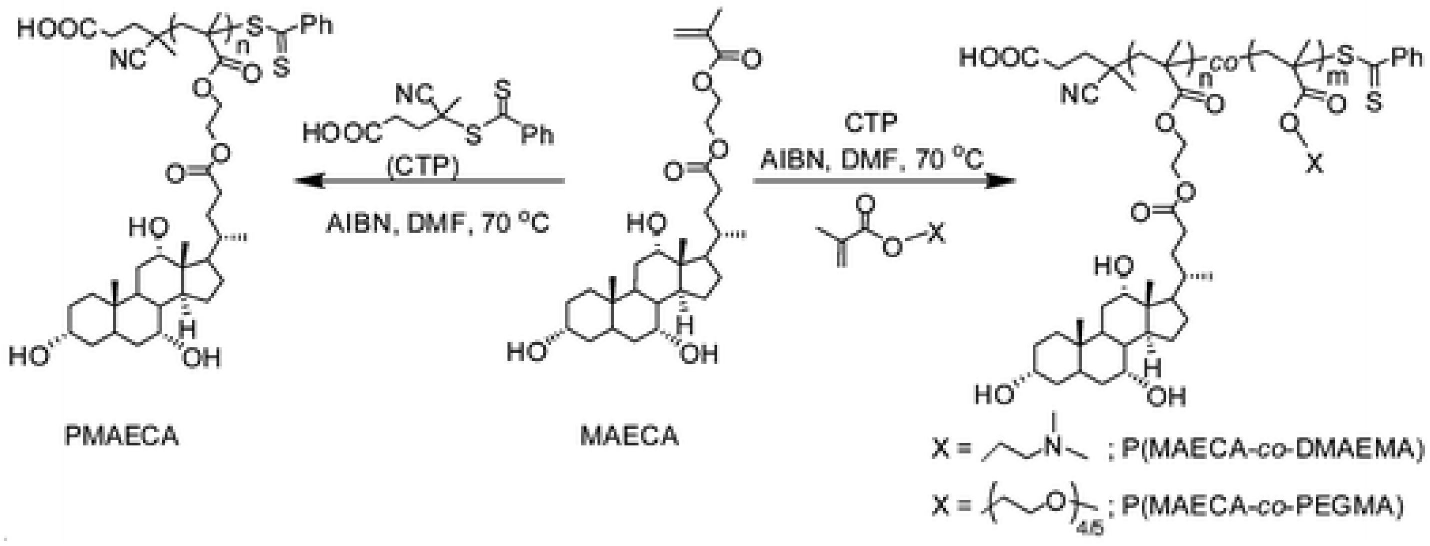
Other polymers?
Reverse phenomenon to LCST?

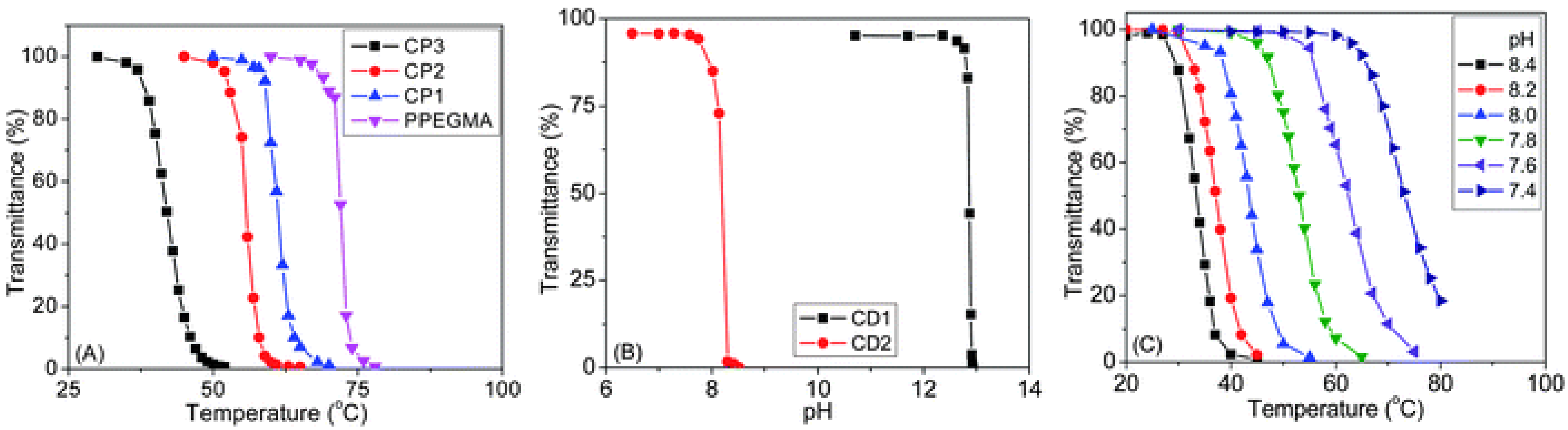
LCST determination



https://www.beilstein-journals.org/bjoc/content/pdf/1860-5397-10-203.pdf

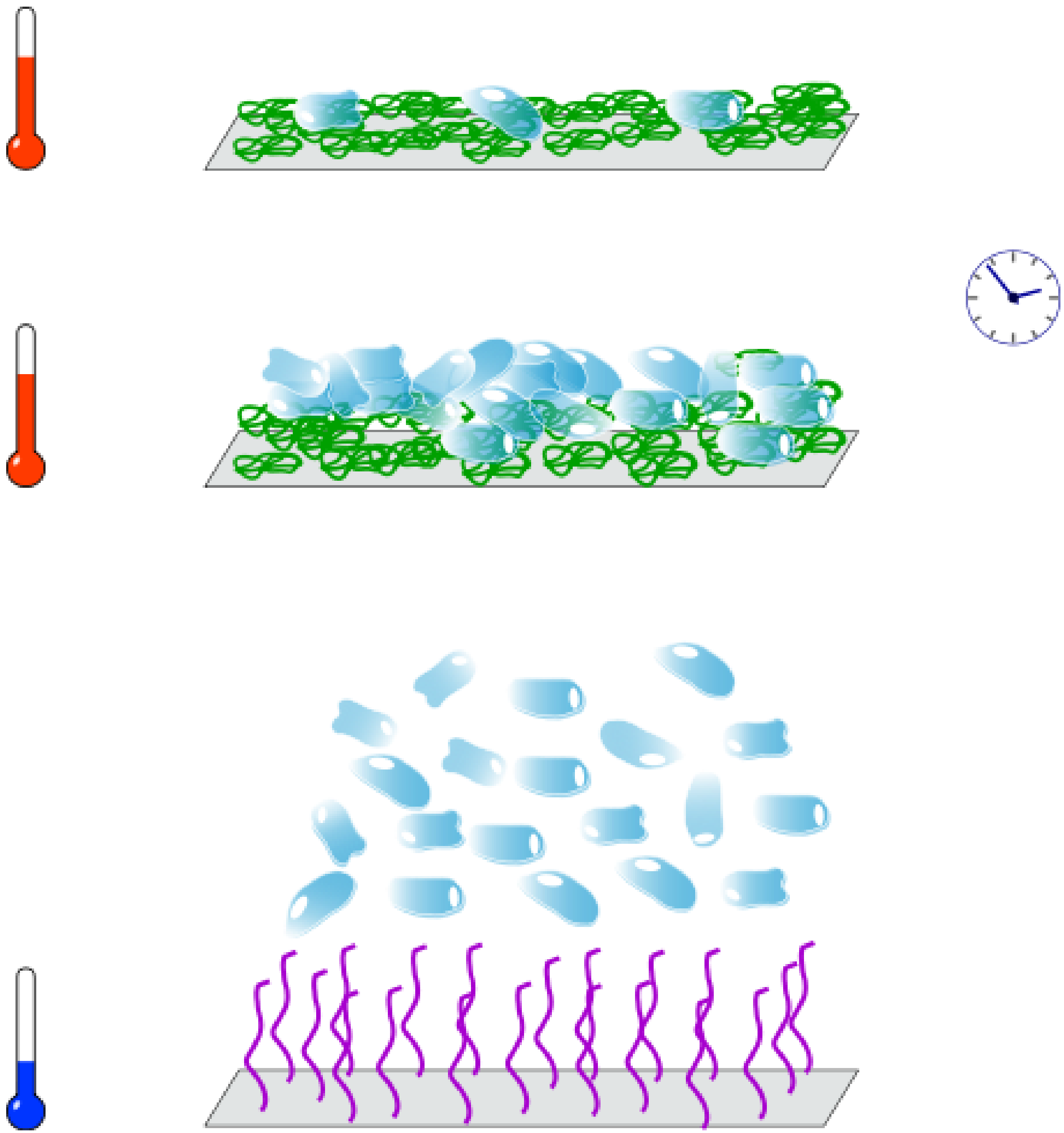
Tunable LCST





http://pubs.rsc.org/en/content/articlehtml/2014/py/c3py01317k

LCST polymer for cell attachment/detachment



Tissue engineering

For some polymers it was demonstrated that thermoresponsive behavior can be transferred to surfaces. The surface is either coated with a polymer film or the polymer chains are bound covalently to the surface. This provides a way to control the wetting properties of a surface by **small temperature changes**. The described behavior can be exploited in tissue engineering since the adhesion of cells is strongly dependent on the hydrophilicity/hydrophobicity.[5][14] This way, it is possible to **detach cells from a cell culture dish by only small changes in temperature**, without the need to additionally use enzymes (see figure).

Learning Resources

www.msm,cam.ac.uk/

University of Cambridge

Department of Materials Science and Metallurgy

Teaching: DoITPoMS Project

Library of Teaching and Learning Packages for Materials Science

www.msm.cam.ac.uk/doitpoms/tlplib/index.php

THE GLASS TRANSITION IN POLYMERS (required reading)

The Macrogalleria

www.psrc.usm.edu/macrog/index.htm

Read through levels 2-5

References

Dr. Sudhir Khetan

1. http://orzo.union.edu/~khetans
2. http://orzo.union.edu/~khetans/Teaching/BNG331/L1%20-%20Introduction%20to%20Biomaterials.pdf
3. https://www.union.edu/search/?q=biomaterials+course#gsc.tab=0&gsc.q=biomaterials%20course&gsc.page=4

Dr. Patrick Tresco

1. http://www.bioen.utah.edu/faculty/pat/Courses/biomaterials/index.htmlAmorphous rubber.
2. http://www.bioen.utah.edu/faculty/pat/Courses/biomaterials/coursenotes.html